RWTH AACHEN UNIVERSITY

Computer Vision – Lecture 15

Local Features

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Computer Vision

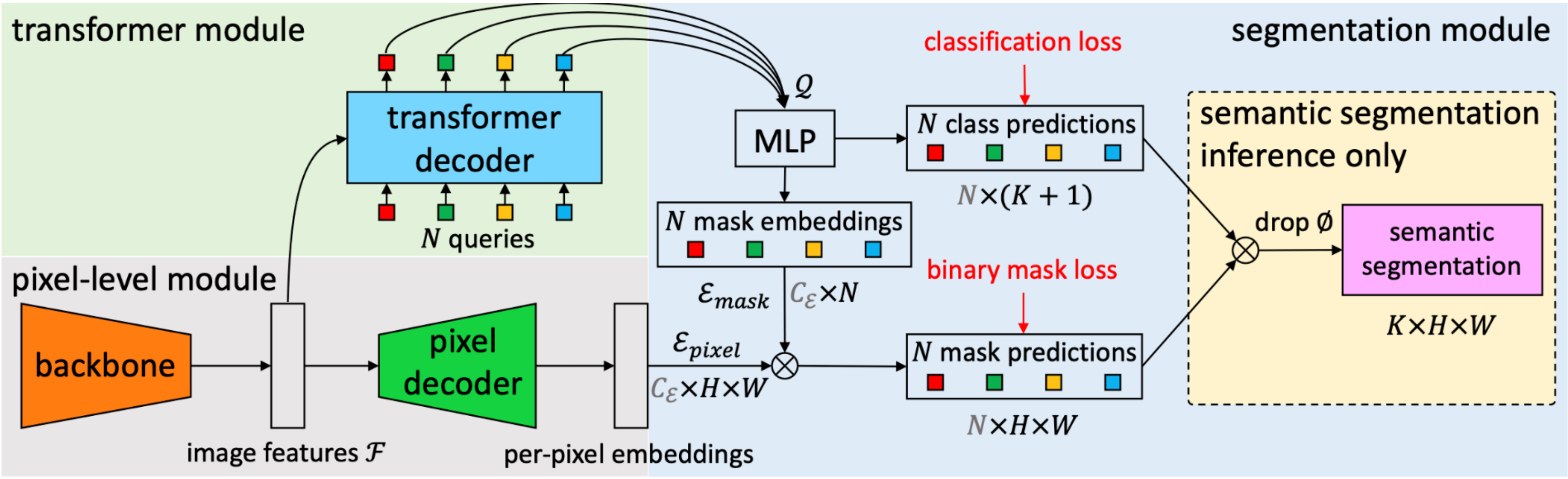
Course Outline

- Image Processing Basics
- Segmentation & Grouping
- Object Recognition & Categorization
- Deep Learning for Vision
- 3D Reconstruction
    - Local Features
    - Epipolar Geometry and Stereo Basics
    - Camera calibration & Triangulation
    - Uncalibrated Reconstruction & Structure-from-Motion

Topics of This Lecture

- Attention & Transformers
  - Motivation
- Transformer Architecture
  - General Attention Layer
  - Self-Attention
  - Positional Encoding
  - Original Transformer Architecture
- Transformers for Vision
  - Image captioning with Transformers
  - Vision Transformers: ViT, Swin
  - Transformer Architectures

Transformers for Dense Prediction: MaskFormer

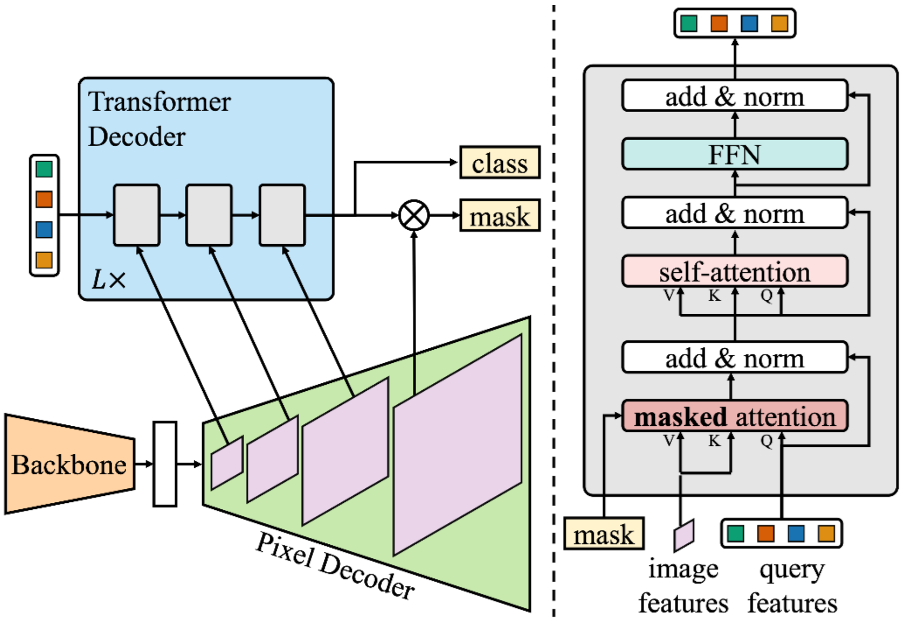


- MaskFormer ideas
  - Prior work: Segmentation considered as a per-pixel classification problem.
  - *Here*: Instead consider it as a mask classification problem.
    - Predict a set of binary masks
    - Each mask is associated with a single global class label

$\Rightarrow$ Unifies *semantic segmentation*, *instance segmentation*, and *panoptic segmentation*

B. Cheng, I. Misra, A. Schwing, A. Kirillov, A. Girdhar, Masked-Attention Mask Transformer for Universal Image Segmentation, CVPR 2022.

Extension: Mask2Former

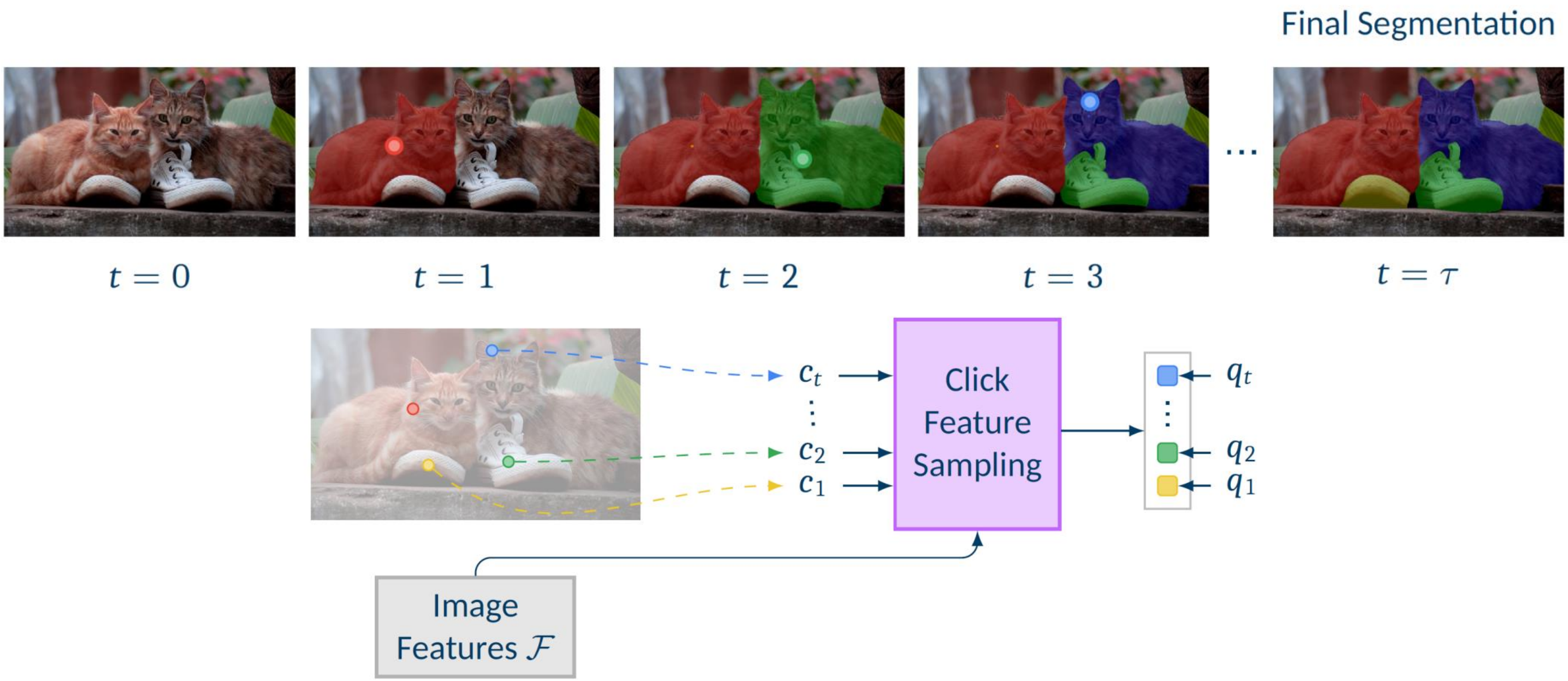


- Mask2Former
  - Same meta architecture as MaskFormer
  - Transformer decoder now contains a masked attention operator
    - Extracts more localized features by constraining cross-attention to within the foreground region of the predicted mask
  - Multi-scale processing using pixel decoder's feature pyramid
    - Better handling of small objects

Example: Interactive Segmentation

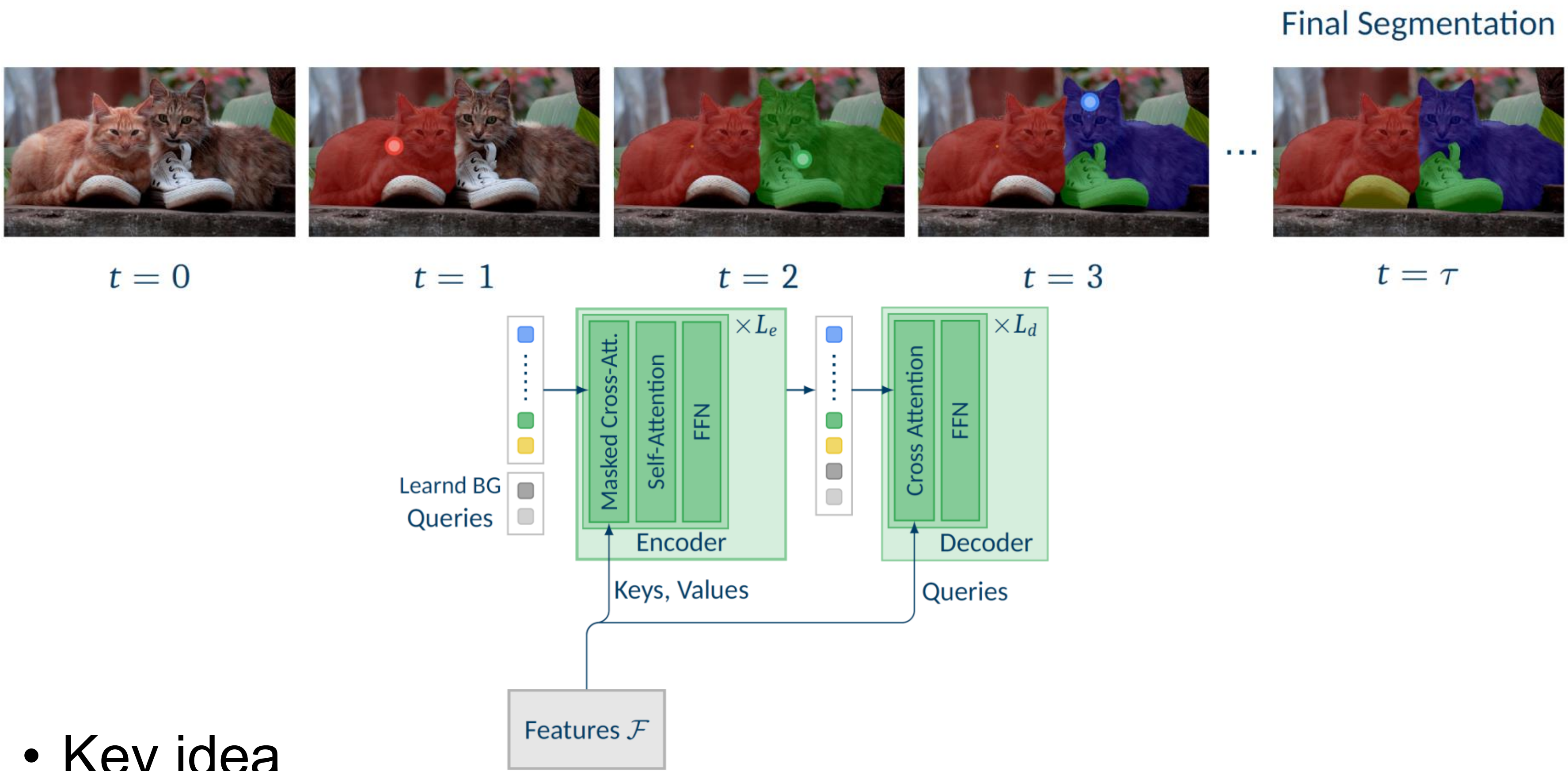
[Mahadevan, Rana, Hermans, Leibe, ICCV'23]

Clicks as Spatiotemporal Sequence



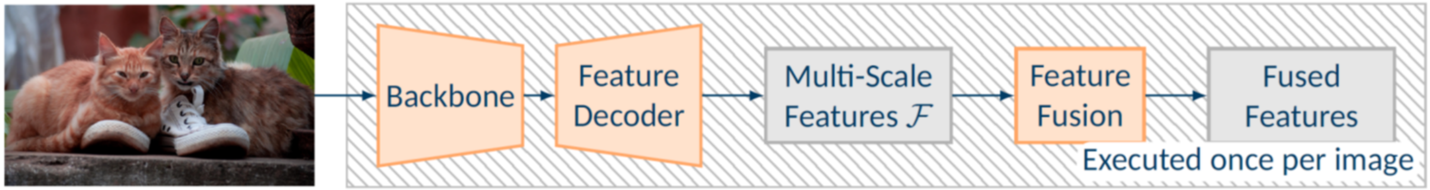
- Key idea
  – Reformulate clicks as spatiotemporal queries

Clicks as Spatiotemporal Sequence



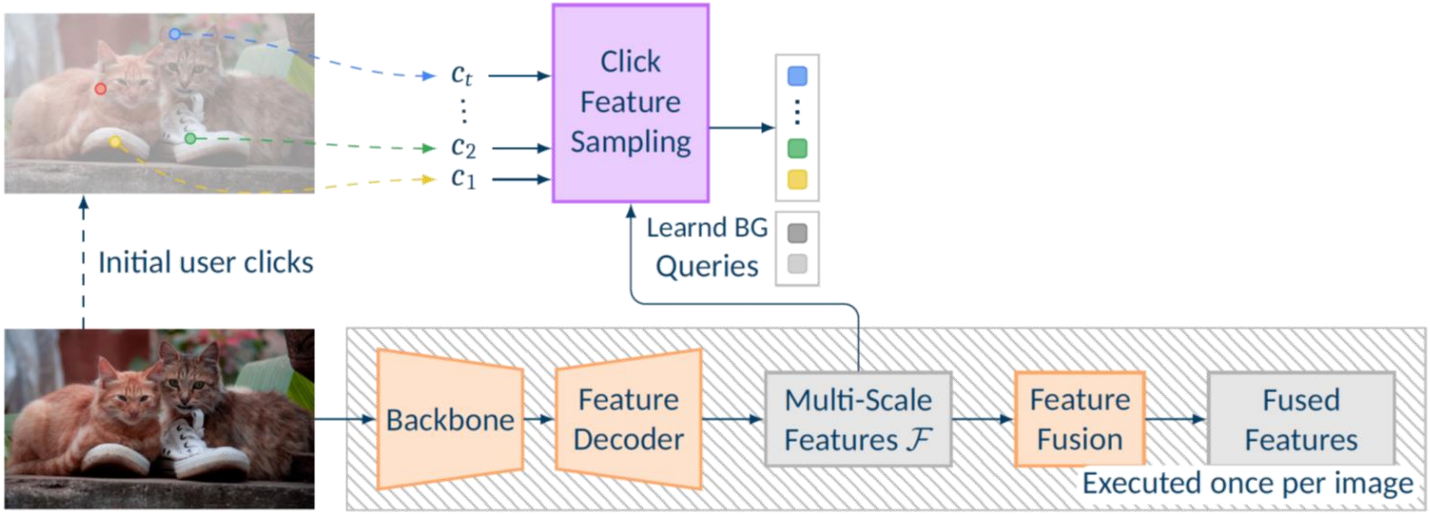
- Key idea
  - Reformulate clicks as spatiotemporal queries processed by a Mask Transformer module

DynaMITe Architecture



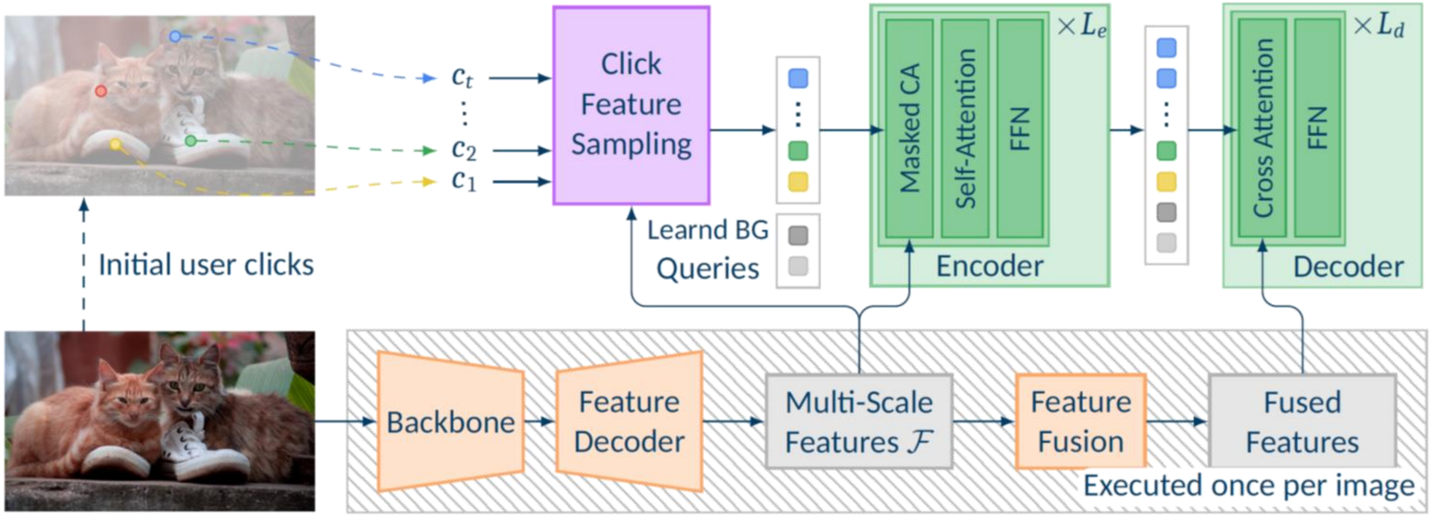
[Mahadevan, Rana, Hermans, Leibe, ICCV'23]

DynaMITe Architecture



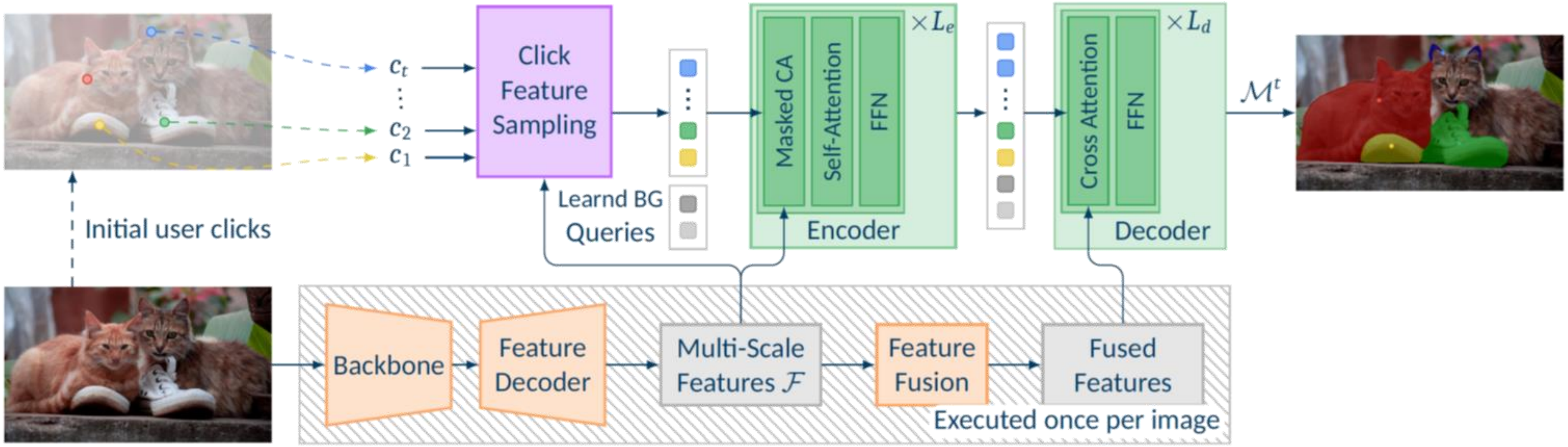
[Mahadevan, Rana, Hermans, Leibe, ICCV'23]

DynaMITe Architecture



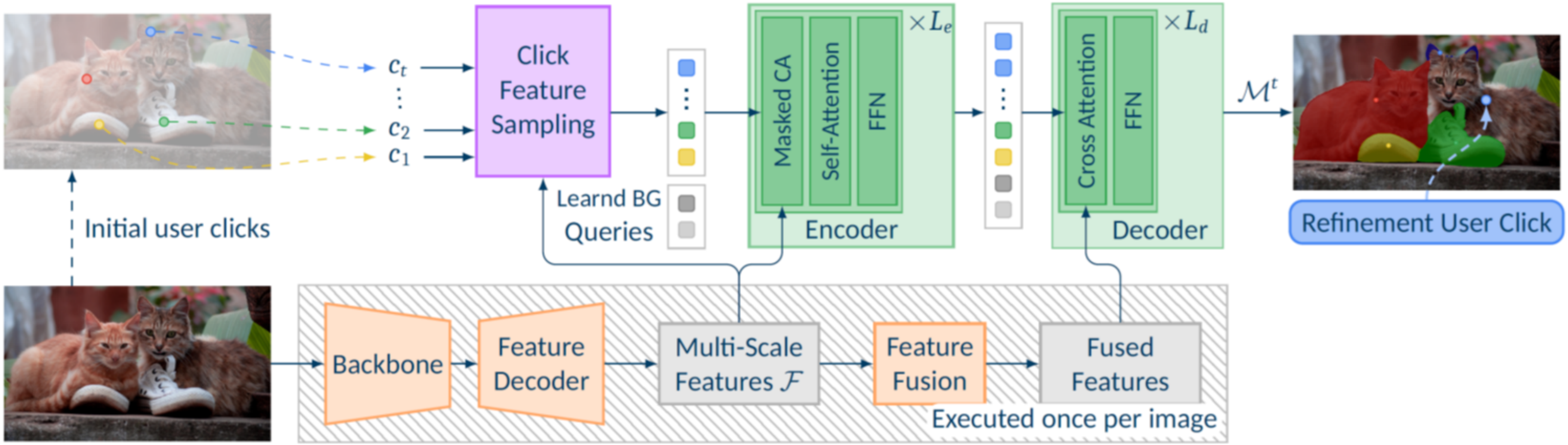
[Mahadevan, Rana, Hermans, Leibe, ICCV'23]

DynaMITe Architecture



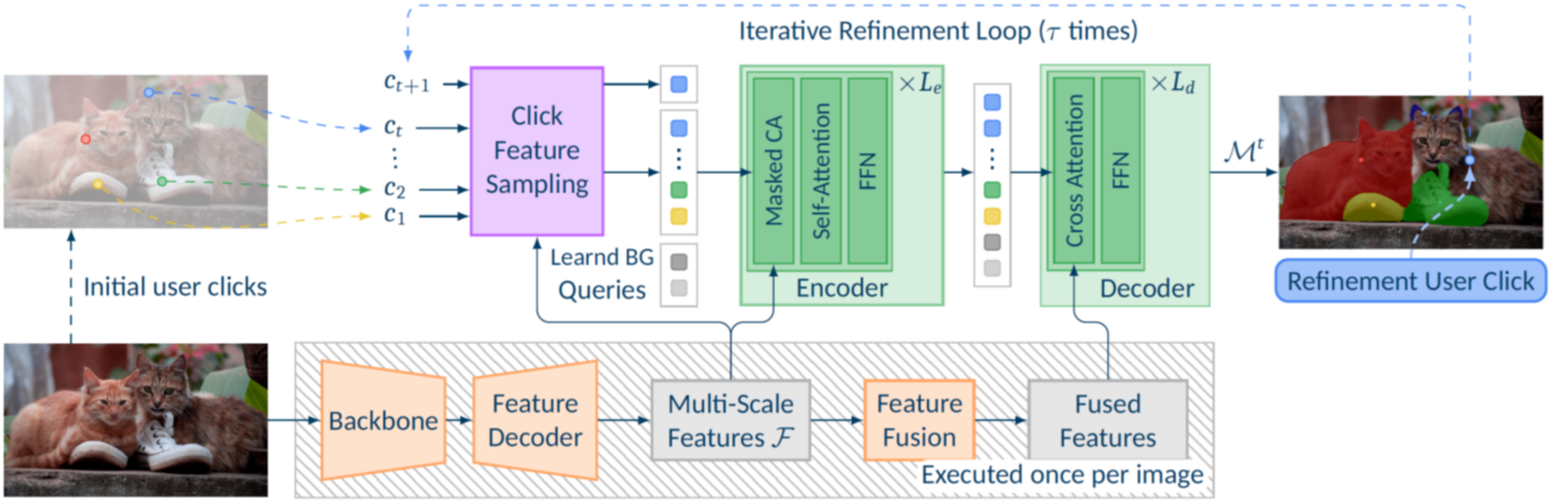
[Mahadevan, Rana, Hermans, Leibe, ICCV’23]

DynaMITe Architecture



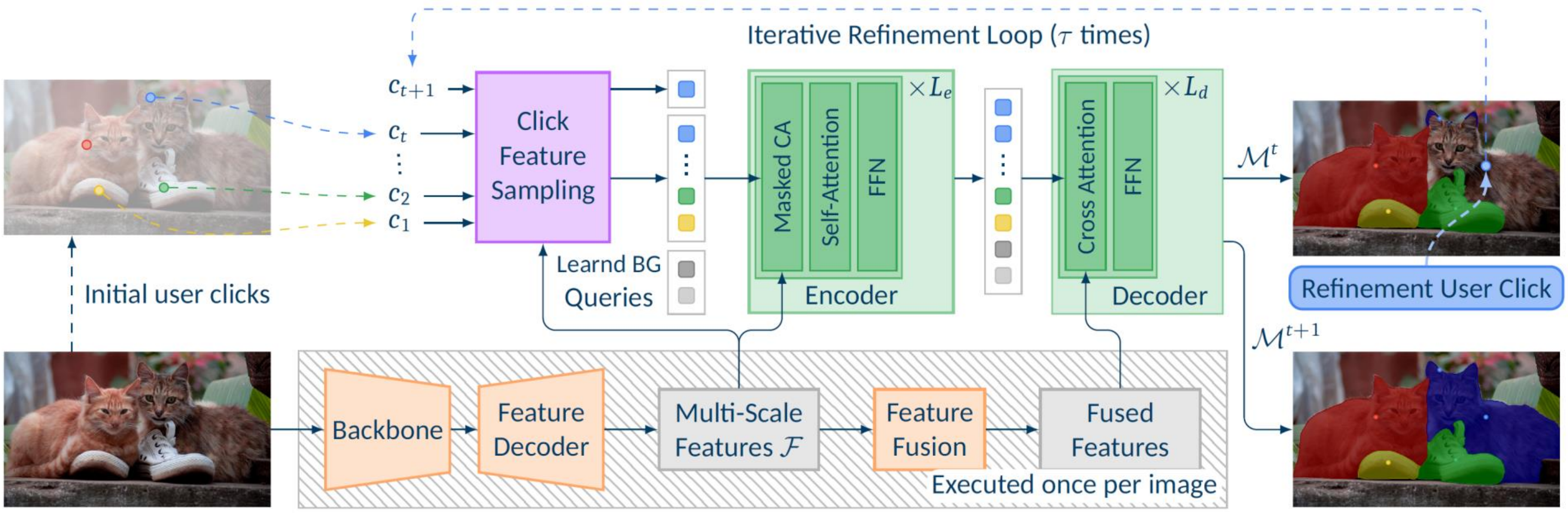
[Mahadevan, Rana, Hermans, Leibe, ICCV'23]

DynaMITe Architecture



[Mahadevan, Rana, Hermans, Leibe, ICCV'23]

DynaMITe Architecture

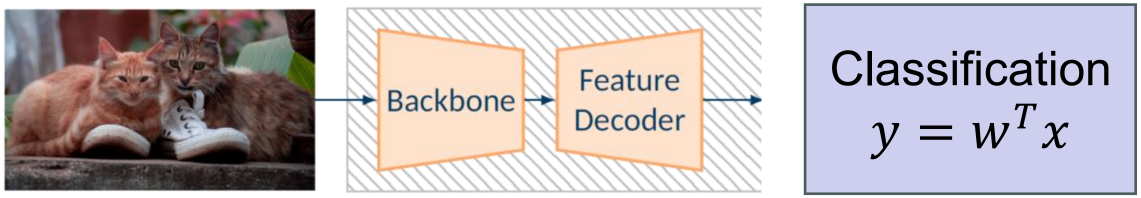


- Advantages
  - Designed for multi-instance segmentation from the start
  - Image-level backbone features need to be computed only once
  - Reduces required #clicks through common background representation

[Mahadevan, Rana, Hermans, Leibe, ICCV'23]

Why Does This Work?

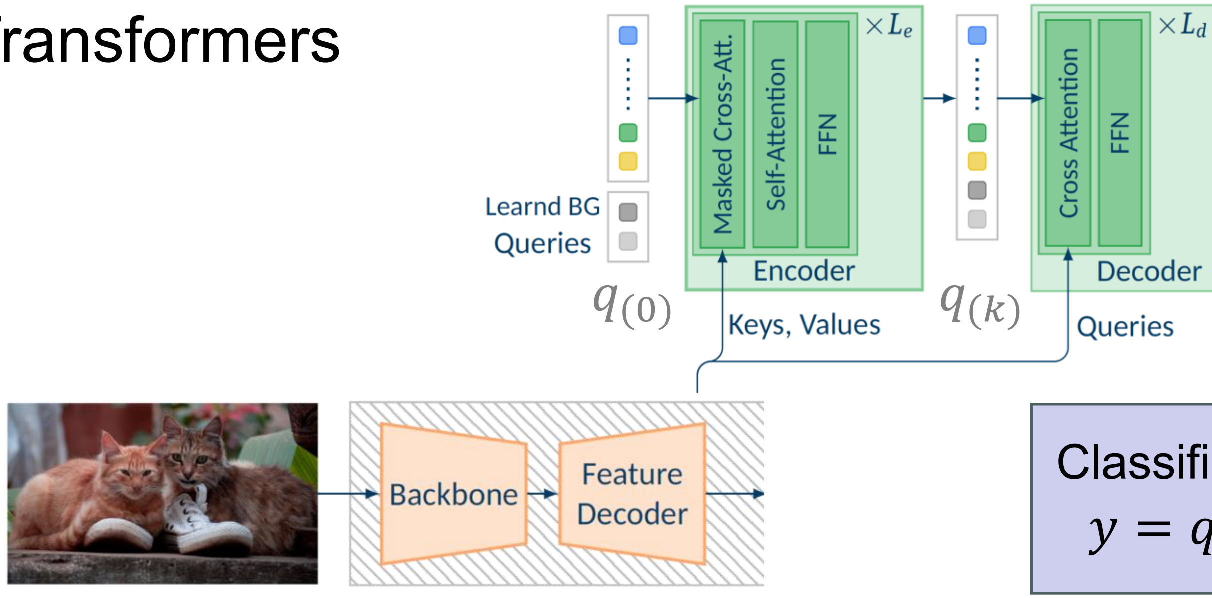
- Traditional CNNs



- What they do
  - Build a representation w for the concept "cat" and compare all pixel features to this representation.
  - w is fixed at training time

Why Does This Work?

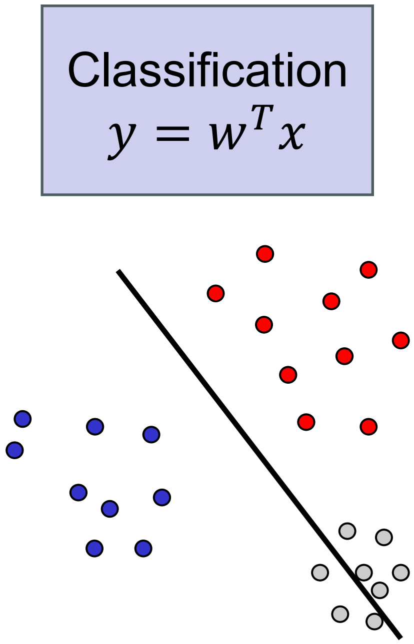
- Mask Transformers

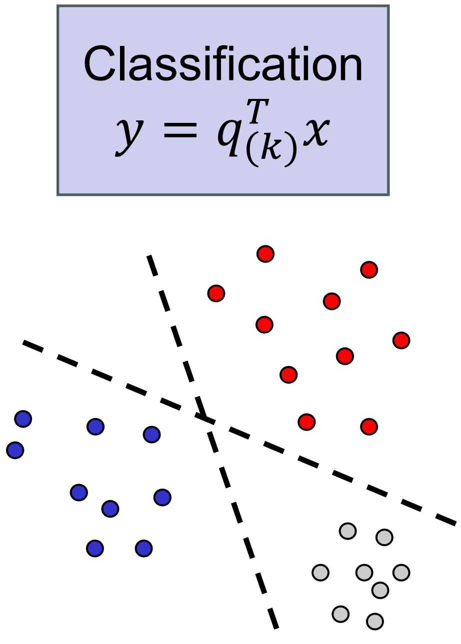


Classification
$y = q_{(k)}^T x$

- What they do
  - Start from an initial query $q_{(0)}$ and successively adapt it $q_{(0)} \longrightarrow q_{(k)}$ *to the specific image at test time*, taking into account the other queries
  - They thus ask "What does it mean to 'segment the brown cat' *in the context of this image?*"

Comparison





- Classic CNNs
  - Decision boundary is fixed at training time
  - Significant training effort needed to get it "always right"
- Mask Transformers
  - Decision boundary can be adjusted at test time, based on context
  - Some indication that this also helps regularize learning

Topics of This Lecture

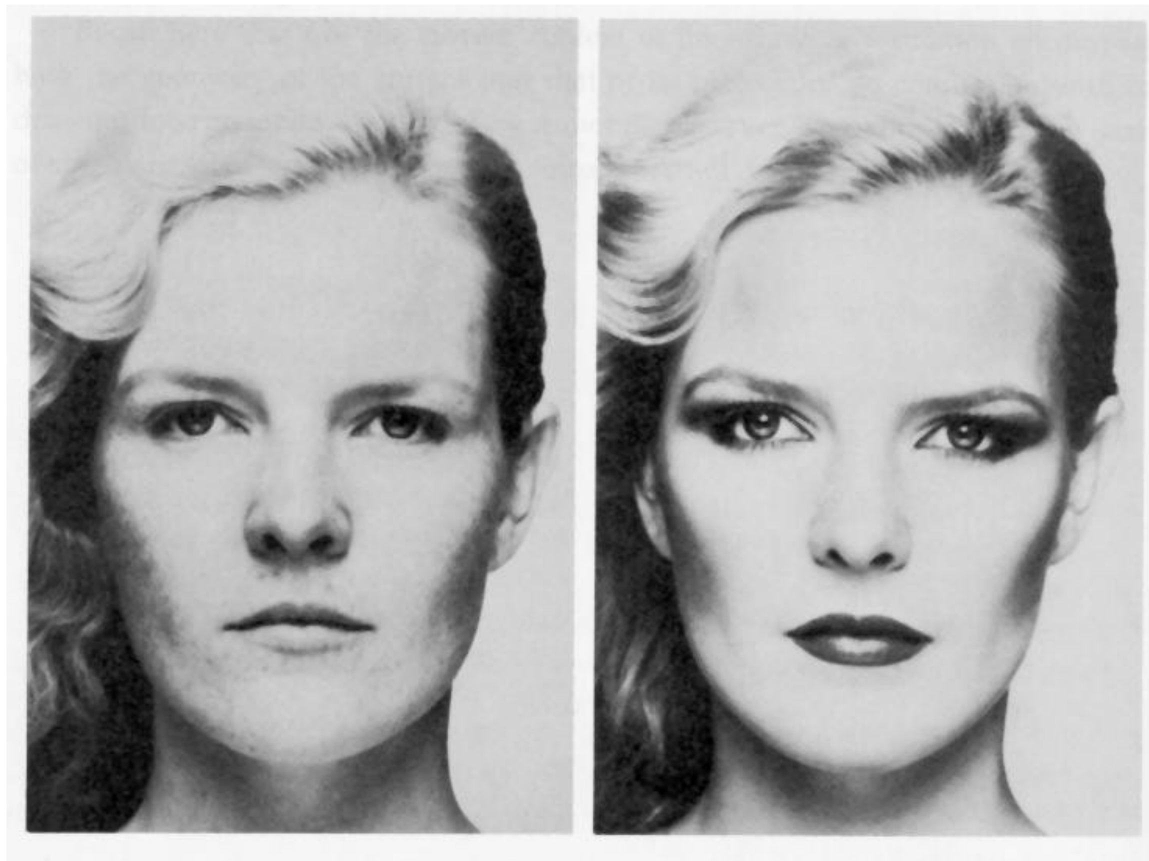
- 3D Reconstruction
  - Visual cues
  - Stereo vision
- Local Invariant Features
  - Motivation
  - Requirements, Invariances
- Keypoint Localization
  - Harris detector
- Local Descriptors
  - Orientation normalization
  - SIFT descriptor

3D Reconstruction

- Goal: Recovery of 3D structure
  - What cues in the image allow us to do this?

Visual Cues

- Shading



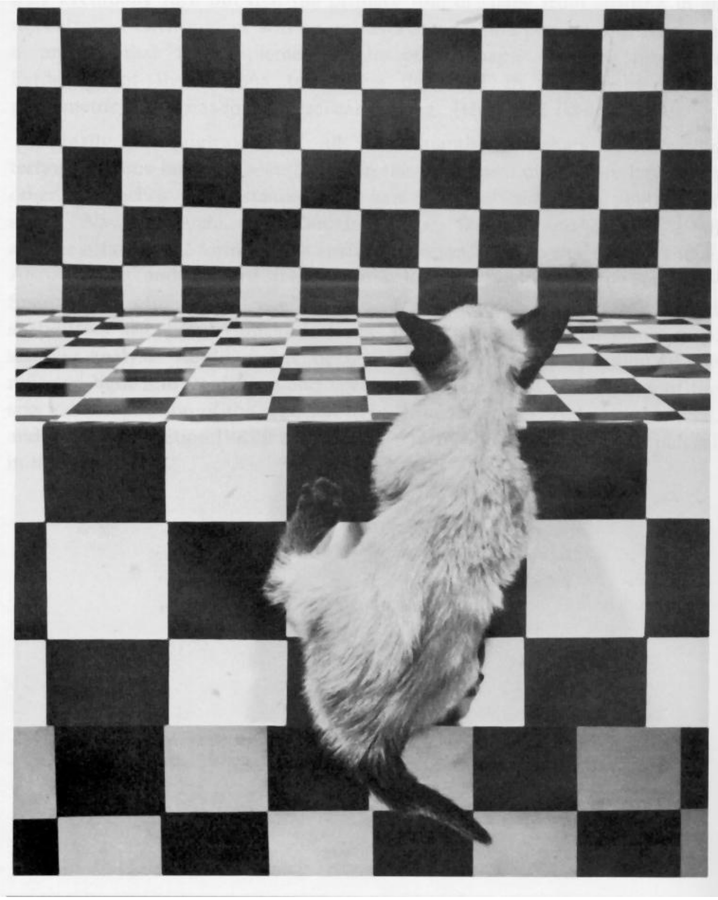
Merle Norman Cosmetics, Los Angeles

Slide credit: Steve Seitz

Visual Cues

- Shading

- Texture



The Visual Cliff, by William Vandivert, 1960

Slide credit: Steve Seitz

Visual Cues

- Shading
- Texture
- Focus

From *The Art of Photography*, Canon

Slide credit: Steve Seitz

Visual Cues

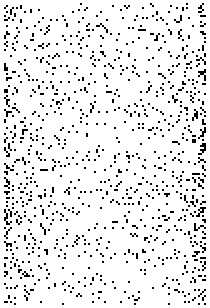
- Shading
- Texture
- Focus
- Perspective

Slide credit: Steve Seitz

Visual Cues

- Shading
- Texture
- Focus
- Perspective
- Motion

Figures from L. Zhang

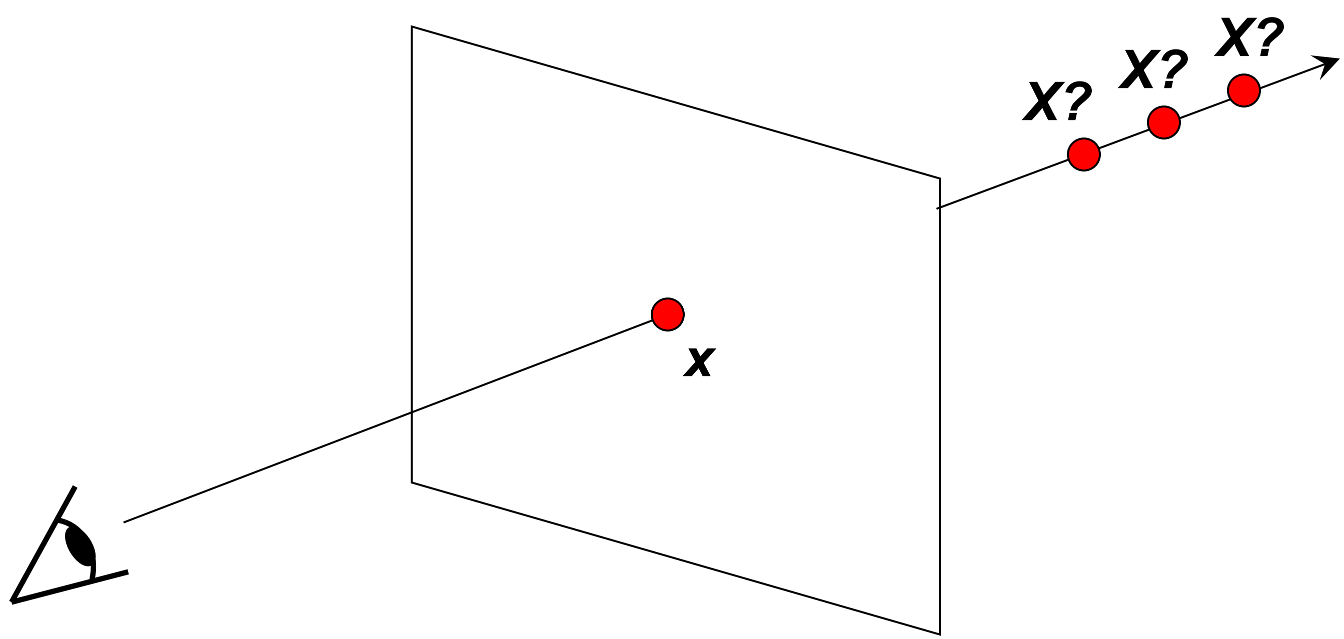


Slide credit: Steve Seitz, Kristen Grauman

http://www.brainconnection.com/teasers/?main=illusion/motion-shape

Our Goal: Recovery of 3D Structure

- We will focus on perspective and motion
- We need *multi-view geometry* because recovery of structure from one image is inherently ambiguous



Slide credit: Svetlana Lazebnik

To Illustrate This Point…

- Structure and depth are inherently ambiguous from single views.

Slide credit: Svetlana Lazebnik, Kristen Grauman

Stereo Vision

http://www.well.com/~jimg/stereo/stereo_list.html

Slide credit: Kristen Grauman

What Is Stereo Vision?

- Generic problem formulation: given several images of the same object or scene, compute a representation of its 3D shape

Slide credit: Svetlana Lazebnik, Steve Seitz

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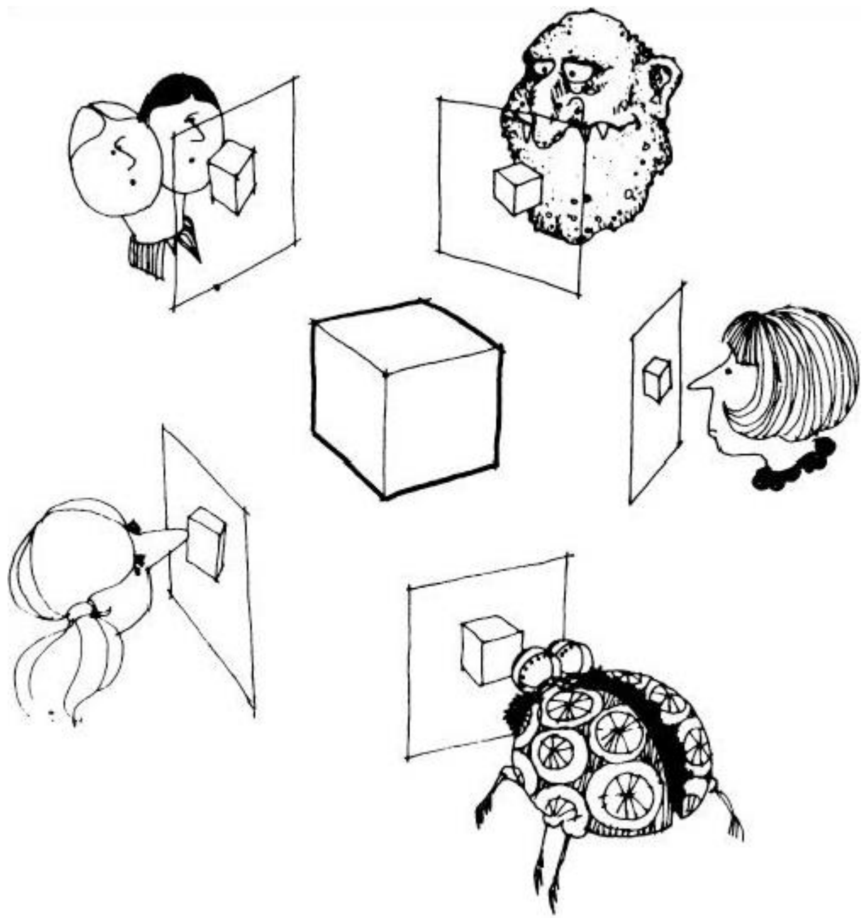


Slide credit: Svetlana Lazebnik, Steve Seitz

What Is Stereo Vision?

- Narrower formulation: given a calibrated binocular stereo pair, fuse it to produce a depth image

Image 1

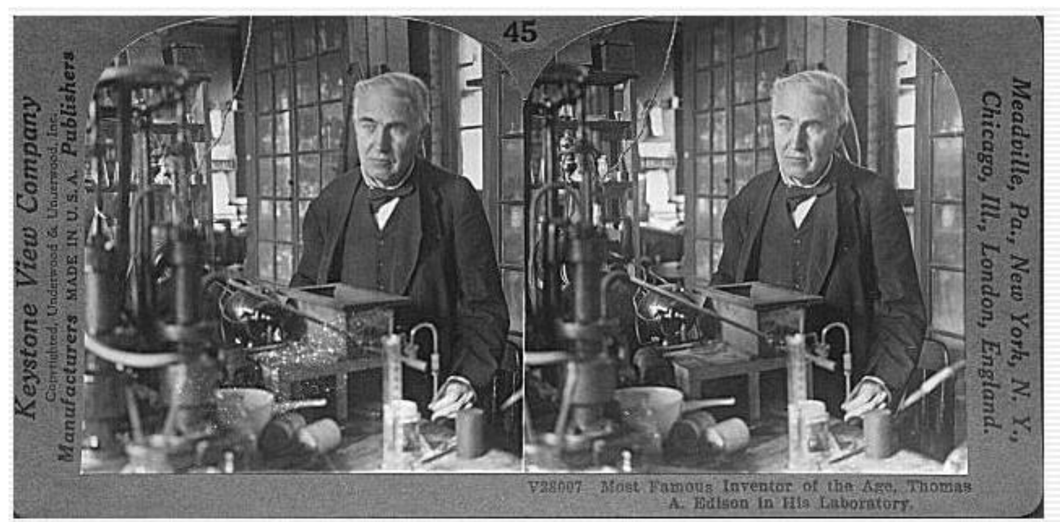
Image 2

Dense depth map

Slide credit: Svetlana Lazebnik, Steve Seitz

What Is Stereo Vision?

- Narrower formulation: given a calibrated binocular stereo pair, fuse it to produce a depth image.
  - Humans can do it



Stereograms: Invented by Sir Charles Wheatstone, 1838

Slide credit: Svetlana Lazebnik, Steve Seitz

What Is Stereo Vision?

- Narrower formulation: given a calibrated binocular stereo pair, fuse it to produce a depth image.
  - Humans can do it

Autostereograms: http://www.magiceye.com

Slide credit: Svetlana Lazebnik, Steve Seitz

What Is Stereo Vision?

- Narrower formulation: given a calibrated binocular stereo pair, fuse it to produce a depth image.
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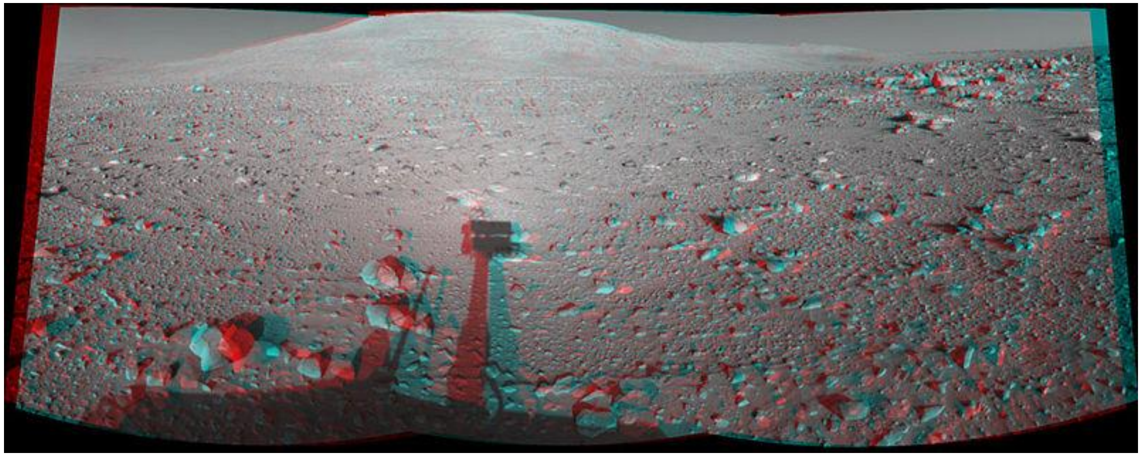
Autostereograms: http://www.magiceye.com

Slide credit: Svetlana Lazebnik, Steve Seitz

Application of Stereo: Robotic Exploration

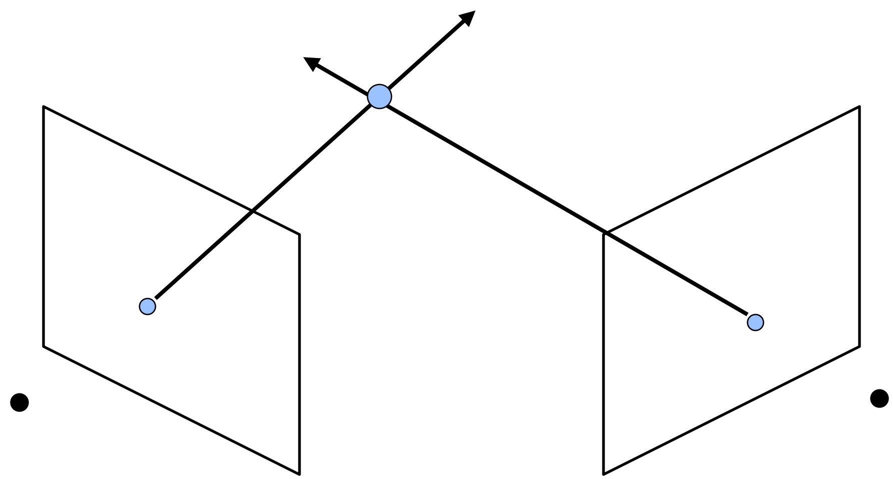
Nomad robot searches for meteorites in Antartica

Real-time stereo on Mars



Slide credit: Steve Seitz

Depth with Stereo: Basic Idea



- Basic Principle: Triangulation
  - Gives reconstruction as intersection of two rays
  - Requires
    - Camera pose (calibration)
    - Point correspondences

Slide credit: Steve Seitz

Topics of This Lecture

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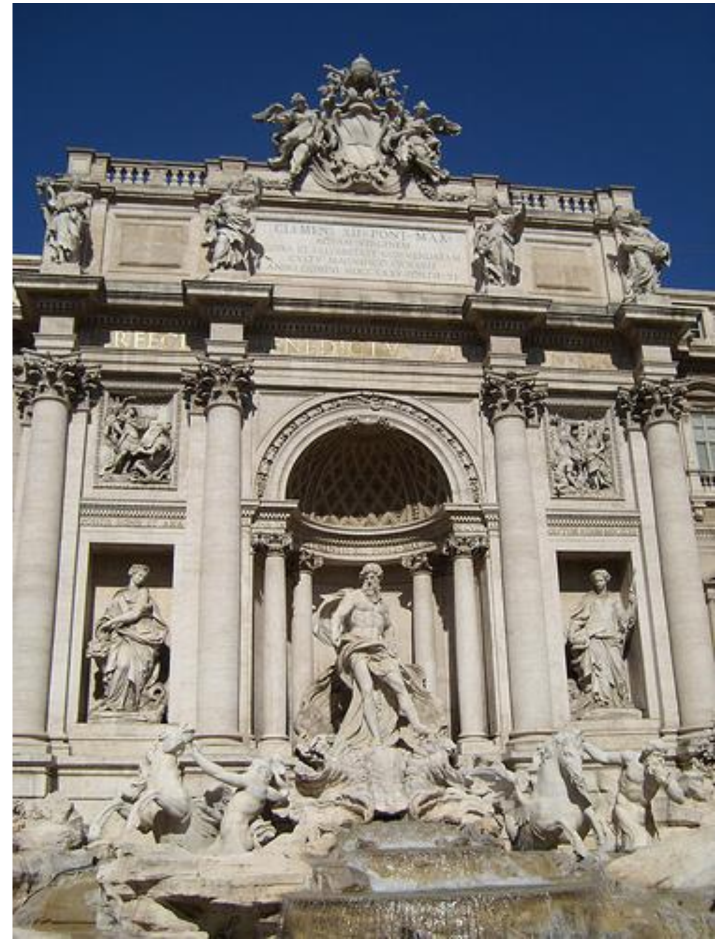
  - Visual cues
  - Stereo vision
- Local Invariant Features
  - Motivation
  - Requirements, Invariances
- Keypoint Localization
  - Harris detector
- Local Descriptors
  - Orientation normalization
  - SIFT descriptor

Application: Image Matching

by Diva Sian



by swashford

How can we find point correspondences between such images?

Slide credit: Steve Seitz

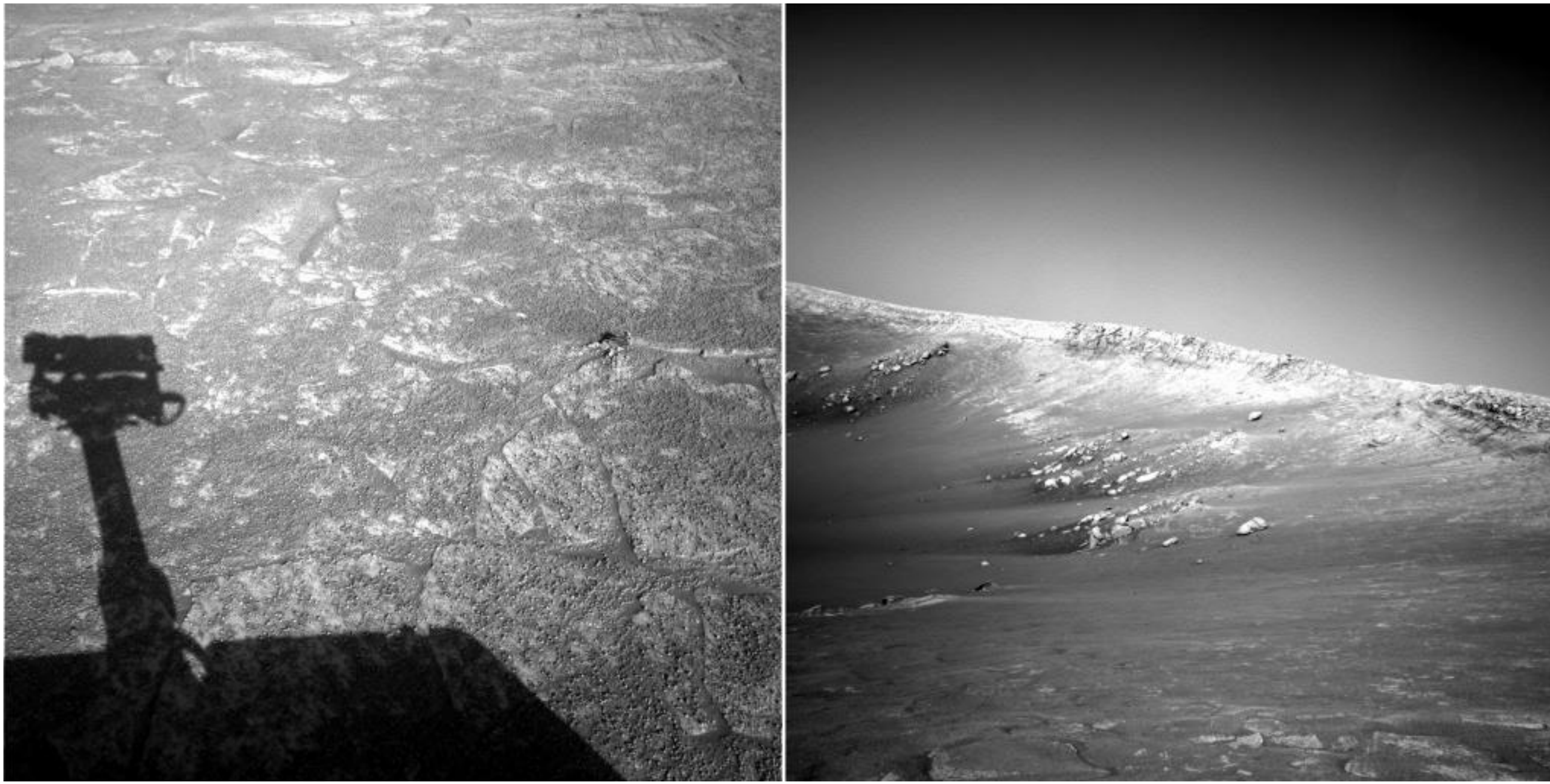
Harder Case

by Diva Sian

by scgbt

Slide credit: Steve Seitz

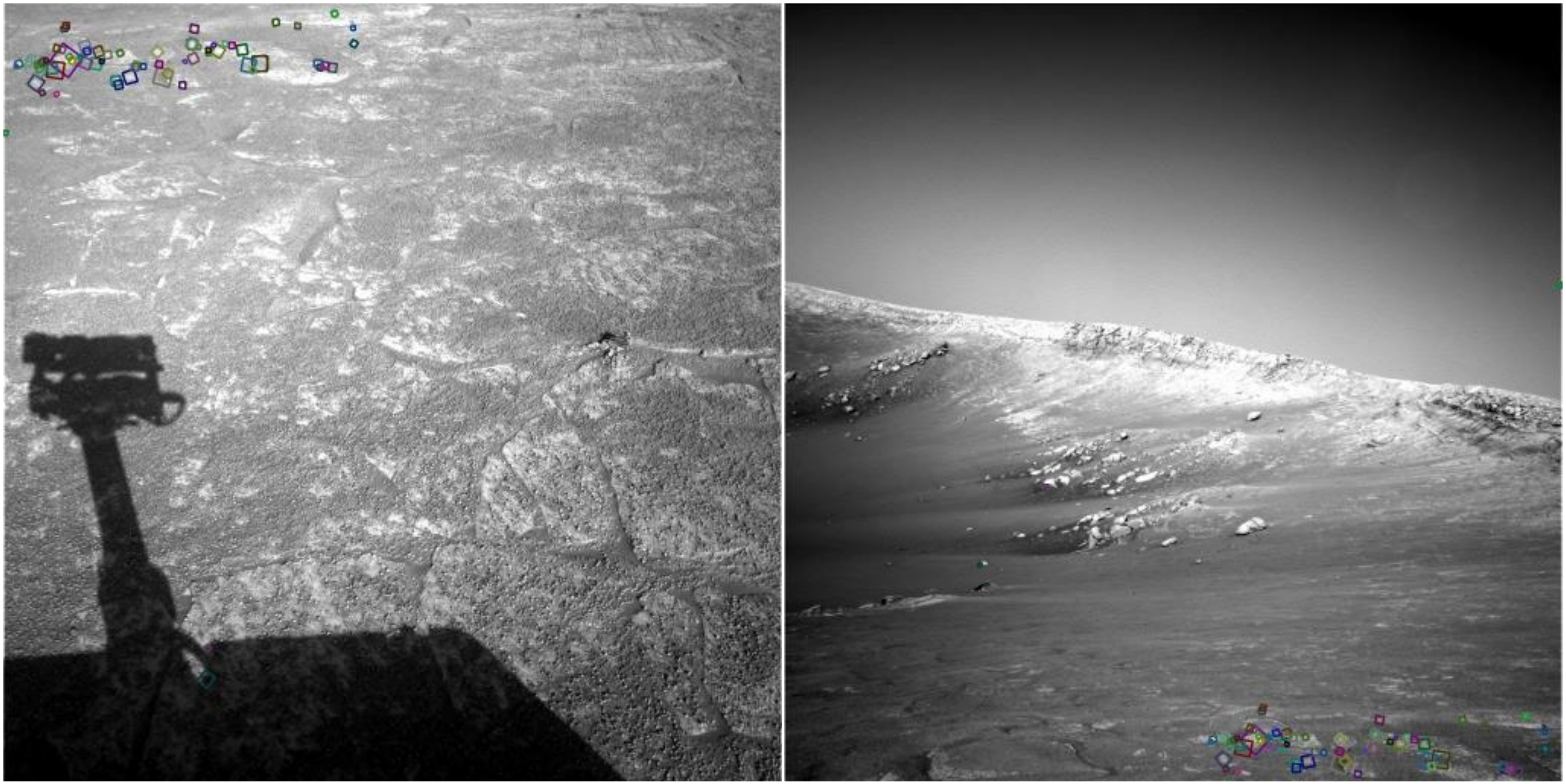
Harder Still?



NASA Mars Rover images

Slide credit: Noah Snavely

Answer Below (Look for Tiny Colored Squares)



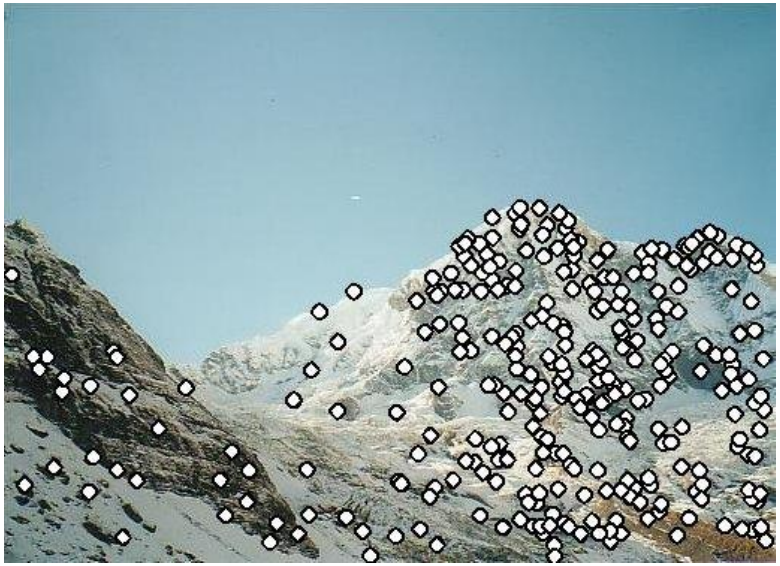
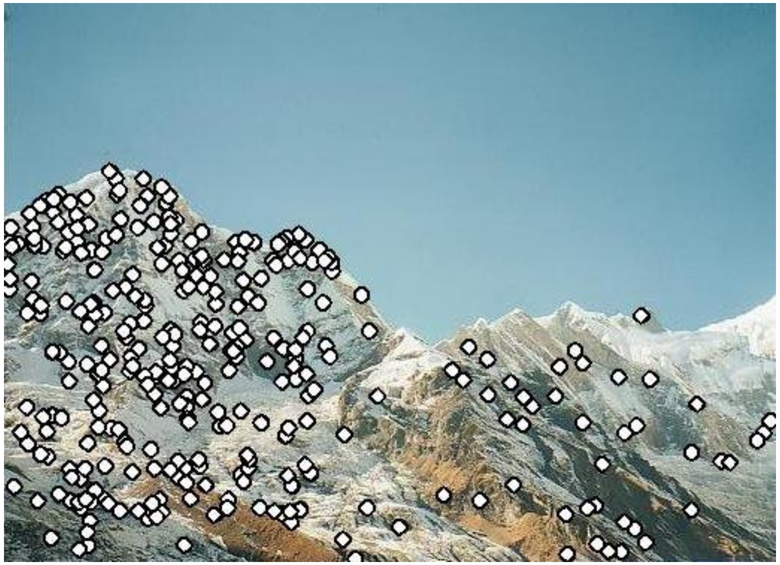
NASA Mars Rover images
with SIFT feature matches

Slide credit: Noah Snavely

Application: Image Stitching


Slide credit: Darya Frolova, Denis Simakov

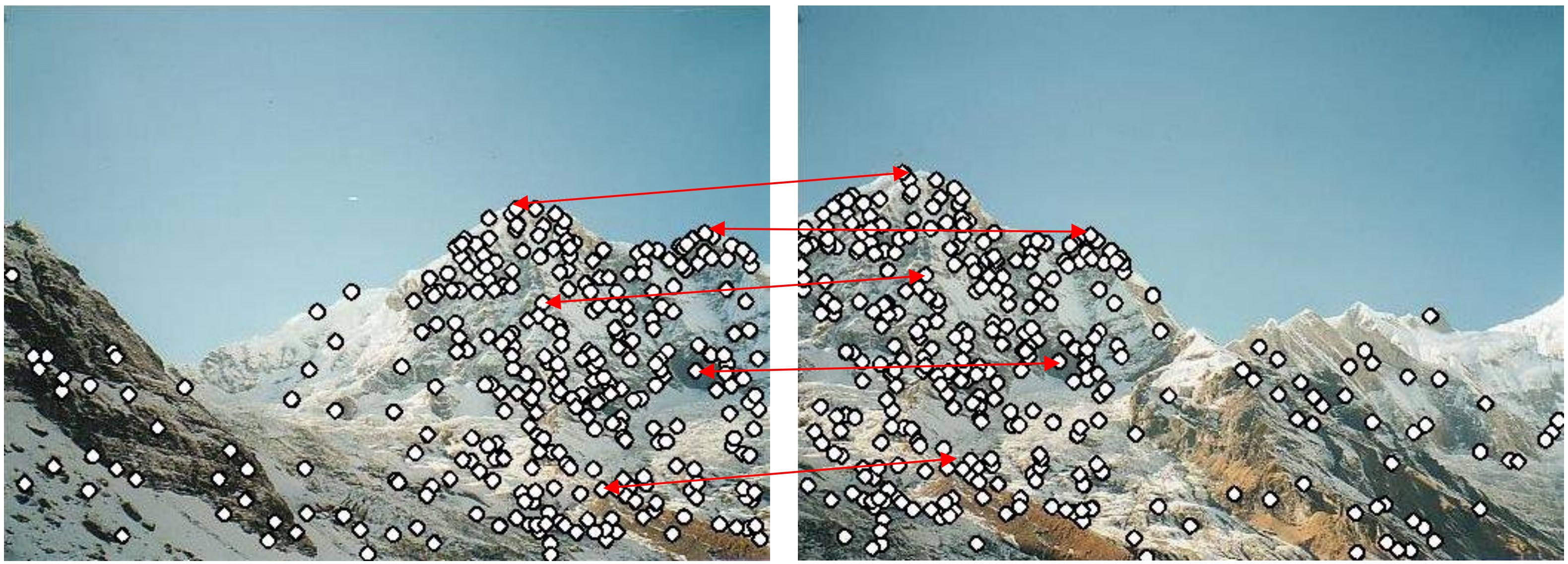
Application: Image Stitching

- Procedure:
  - Detect feature points in both images

Slide credit: Darya Frolova, Denis Simakov

Application: Image Stitching



- Procedure:
  - Detect feature points in both images
  - Find corresponding pairs

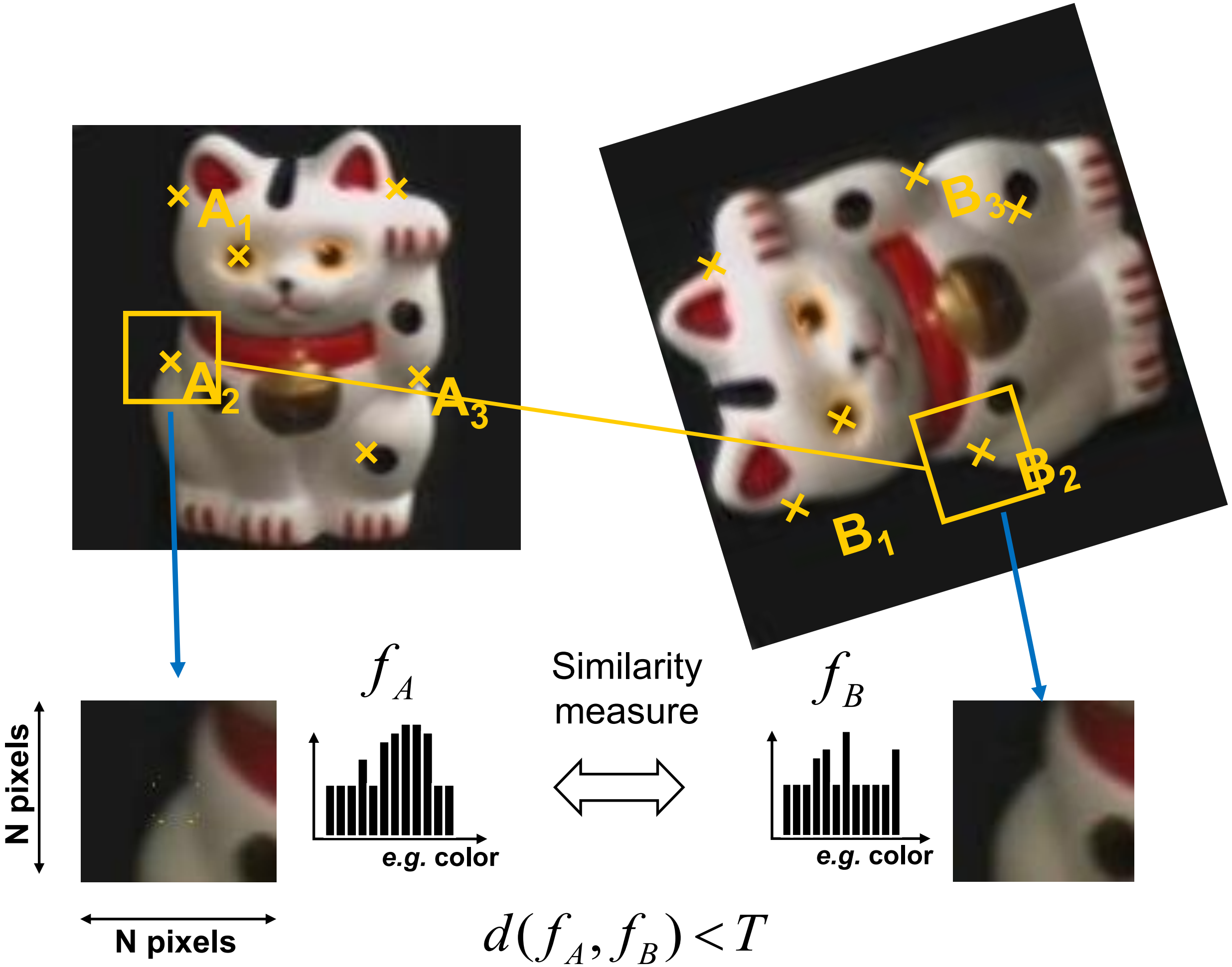
Slide credit: Darya Frolova, Denis Simakov

Application: Image Stitching

- Procedure:
  - Detect feature points in both images
  - Find corresponding pairs
  - Use these pairs to align the images

Slide credit: Darya Frolova, Denis Simakov

General Approach



1. Find a set of distinctive key-points
2. Define a region around each keypoint
3. Extract and normalize the region content
4. Compute a local descriptor from the normalized region
5. Match local descriptors

Common Requirements

- Problem 1:
  - Detect the same point *independently* in both images

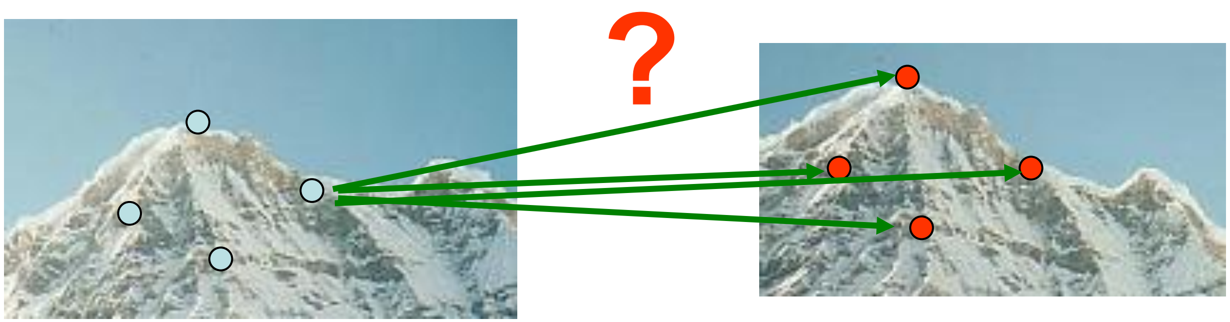
No chance to match!

We need a repeatable detector!

Slide credit: Darya Frolova, Denis Simakov

Common Requirements

- Problem 1:
  - Detect the same point *independently* in both images
- Problem 2:
  - For each point correctly recognize the corresponding one



We need a reliable and distinctive descriptor!

Slide credit: Darya Frolova, Denis Simakov

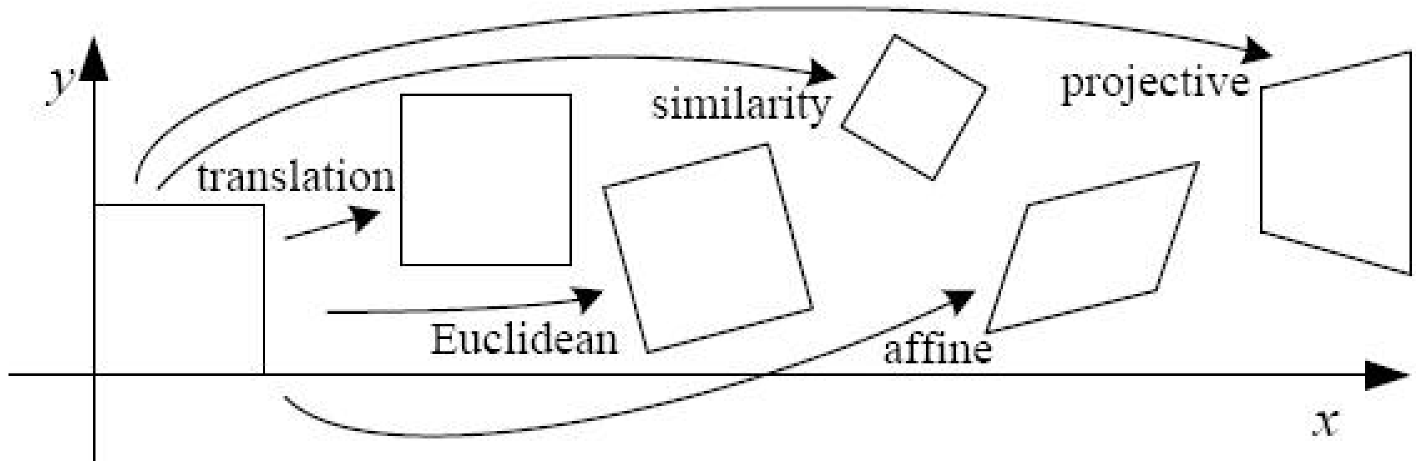
Invariance: Geometric Transformations

- Invariance vs covariance
  - In order for the *region descriptors* to be invariant to a geometric transformation, the *region shape* needs to be covariant to this transformation.

Computer Visionh

Slide credit: Steve Seitz

Levels of Geometric Invariance



Many Existing Detectors Available

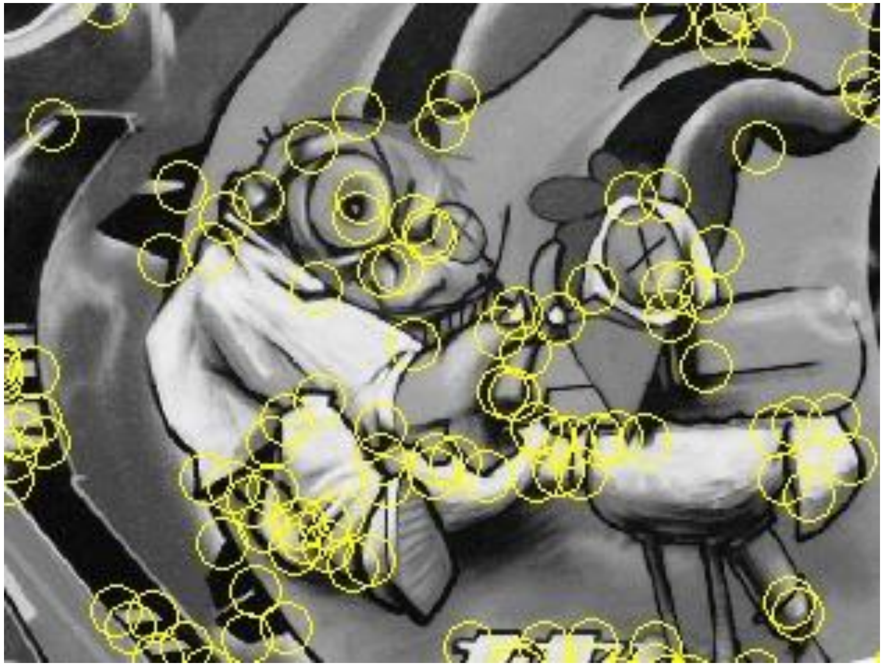
- Hessian & Harris [Beaudet '78], [Harris '88]
- Laplacian, DoG [Lindeberg '98], [Lowe '99]
- Harris-/Hessian-Laplace [Mikolajczyk & Schmid '01]
- Harris-/Hessian-Affine [Mikolajczyk & Schmid '04]
- EBR and IBR [Tuytelaars & Van Gool '04]
- MSER [Matas '02]
- Salient Regions [Kadir & Brady '01]
- Others…

- *Those detectors have become a basic building block for many applications in Computer Vision.*

Topics of This Lecture

- 3D Reconstruction
  - Visual cues
  - Stereo vision
- Local Invariant Features
  - Motivation
  - Requirements, Invariances
- Keypoint Localization
  - Harris detector
- Local Descriptors
  - Orientation normalization
  - SIFT descriptor

Keypoint Localization

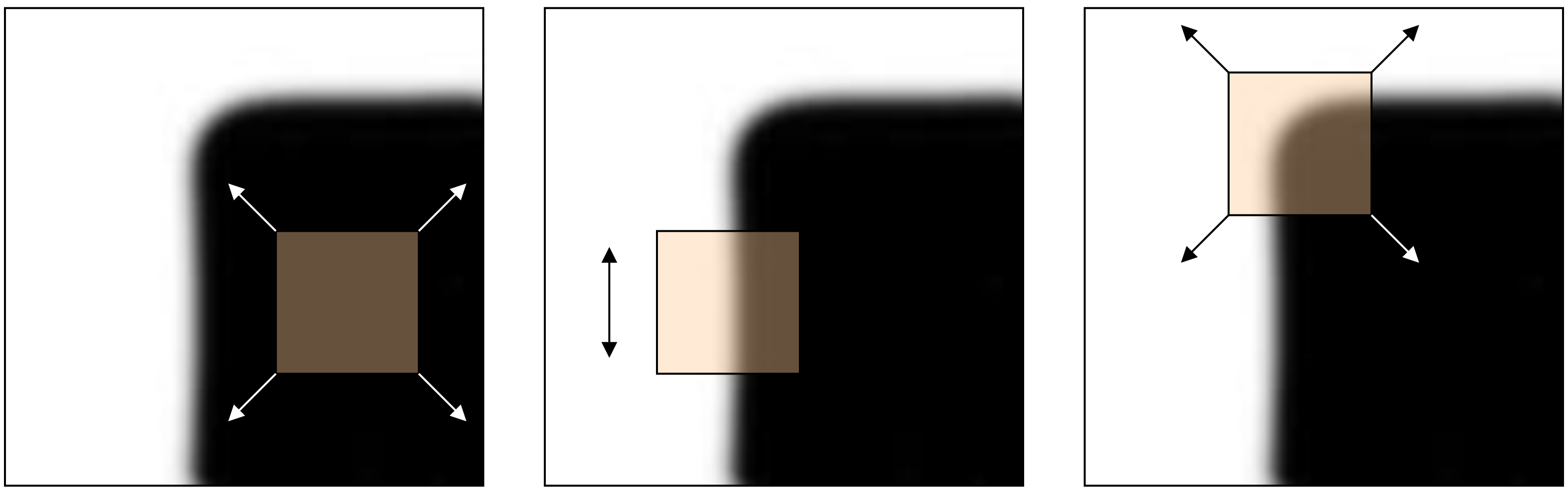


- Goals:
  - Repeatable detection
  - Precise localization
  - Interesting content

$\Rightarrow$ *Look for two-dimensional signal changes*

Corners as Distinctive Interest Points

- Design criteria
  - We should easily recognize the point by looking through a small window (*locality*)
  - Shifting the window in *any direction* should give *a large change* in intensity (*good localization*)



"flat" region:
no change in all directions

"edge":
no change along the edge direction

"corner":
significant change in all directions

Slide credit: Alexej Efros

Harris Detector Formulation

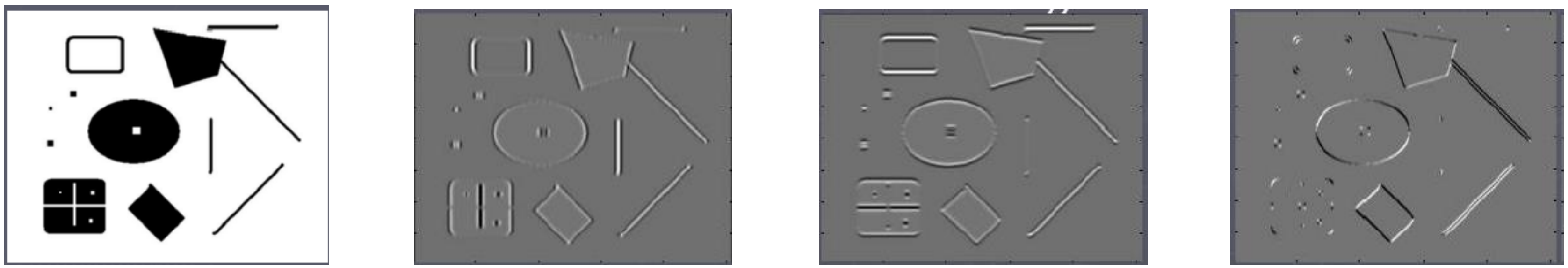


Image I | I_x | I_y | $I_x I_y$

- Start from the second-moment matrix M :

$$M = \sum_{x,y} w(x,y) \begin{bmatrix} I_x^2 & I_x I_y \\ I_x I_y & I_y^2 \end{bmatrix}$$

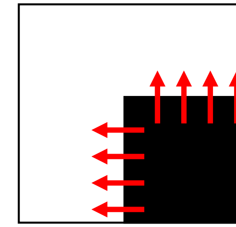
Gradient with respect to x, times gradient with respect to y

Sum over image region – the area we are checking for corner

$$M = \begin{bmatrix} \sum I_x I_x & \sum I_x I_y \\ \sum I_x I_y & \sum I_y I_y \end{bmatrix} = \sum \begin{bmatrix} I_x \\ I_y \end{bmatrix} [I_x \ I_y]$$

Slide credit: Rick Szeliski

What Does This Matrix Reveal?



- First, let's consider an axis-aligned corner:

$$M = \begin{bmatrix} \sum I_x^2 & \sum I_x I_y \\ \sum I_x I_y & \sum I_y^2 \end{bmatrix} = \begin{bmatrix} \lambda_1 & 0 \\ 0 & \lambda_2 \end{bmatrix}$$

- This means:
  - Dominant gradient directions align with x or y axis
  - If either λ is close to 0, then this is not a corner, so look for locations where both are large.

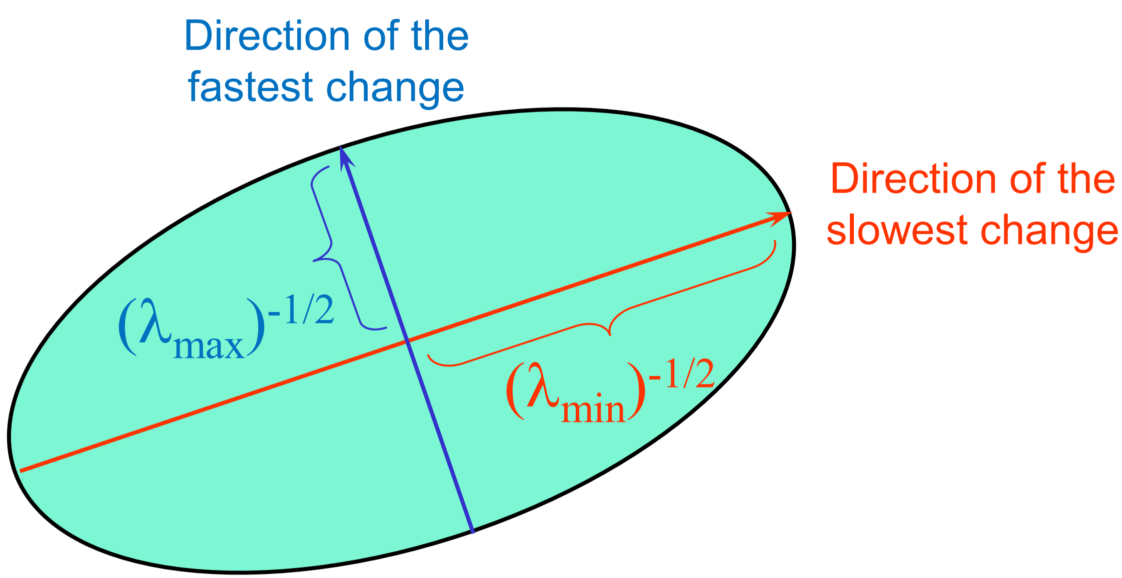
- What if we have a corner that is not aligned with the image axes?

What Does This Matrix Reveal?

- Since M is symmetric, we have $M = R^{-1}\begin{bmatrix}\lambda_1 & 0 \\ 0 & \lambda_2\end{bmatrix}R$

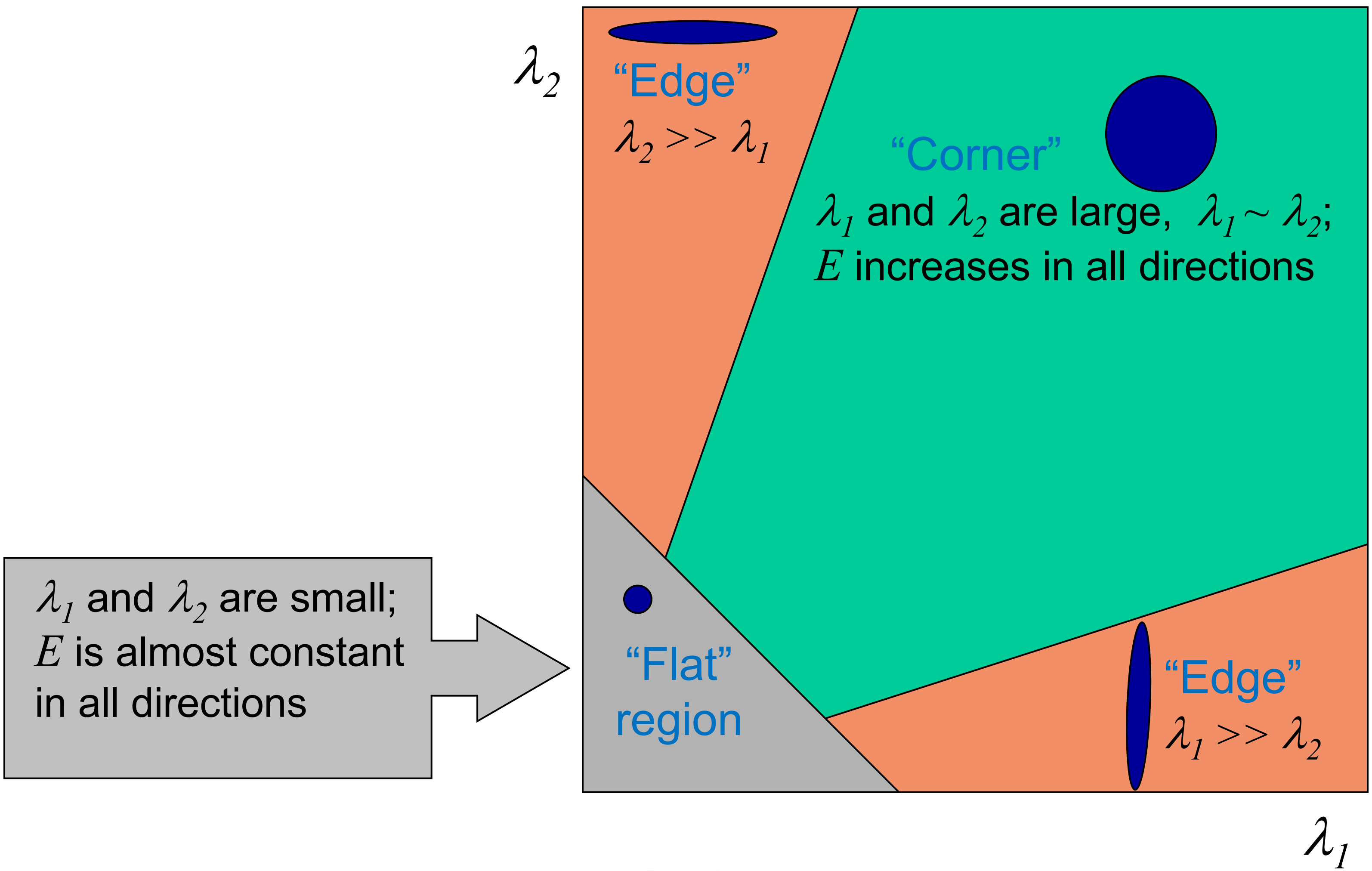
(Eigenvalue decomposition)

- We can visualize M as an ellipse with axis lengths determined by the eigenvalues and orientation determined by R



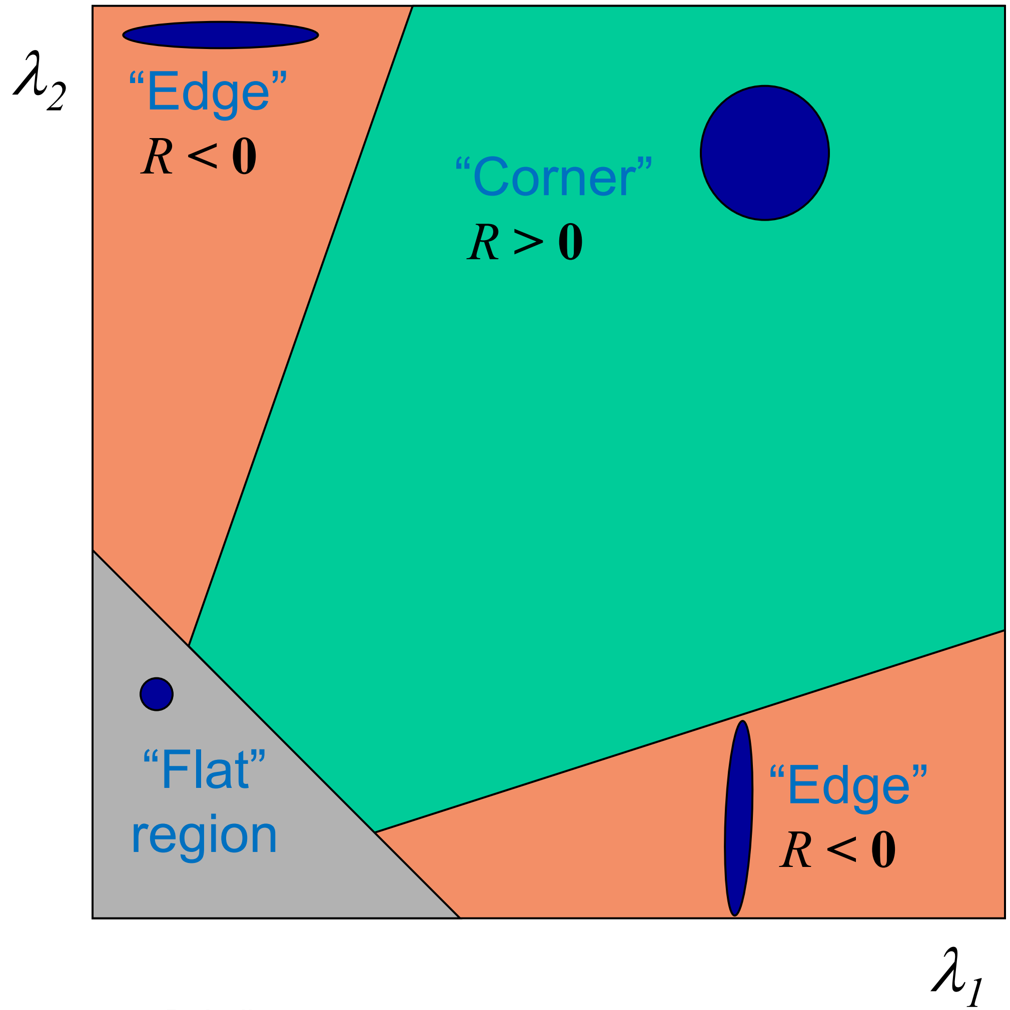
Interpreting the Eigenvalues

- Classification of image points using eigenvalues of M:



Corner Response Function

$$R = \det(M) - \alpha \operatorname{trace}(M)^2 = \lambda_1 \lambda_2 - \alpha(\lambda_1 + \lambda_2)^2$$



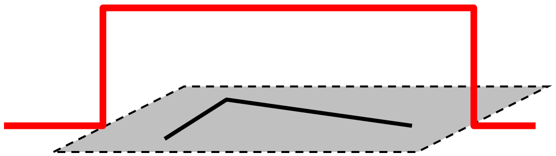
- Fast approximation
  - Avoid computing the eigenvalues
  - α: constant (0.04 to 0.06)

Window Function $w(x,y)$

$$M = \sum_{x,y} w(x,y) \begin{bmatrix} I_x^2 & I_x I_y \\ I_x I_y & I_y^2 \end{bmatrix}$$

- Option 1: uniform window
  - Sum over square window

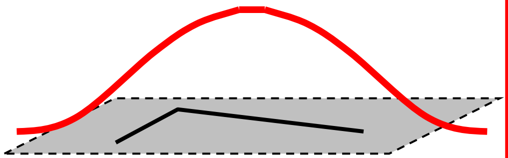
$$M = \sum_{x,y} \begin{bmatrix} I_x^2 & I_x I_y \\ I_x I_y & I_y^2 \end{bmatrix}$$



1 in window, 0 outside

  - Problem: not rotation invariant
- Option 2: Smooth with Gaussian
  - Gaussian already performs weighted sum

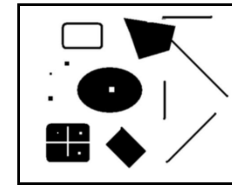
$$M = g(\sigma) * \begin{bmatrix} I_x^2 & I_x I_y \\ I_x I_y & I_y^2 \end{bmatrix}$$



Gaussian

  - Result is rotation invariant

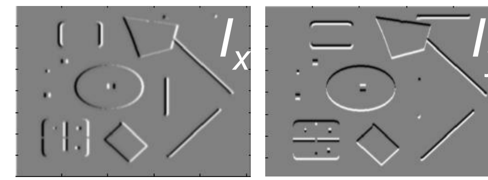
Summary: Harris Detector [Harris88]



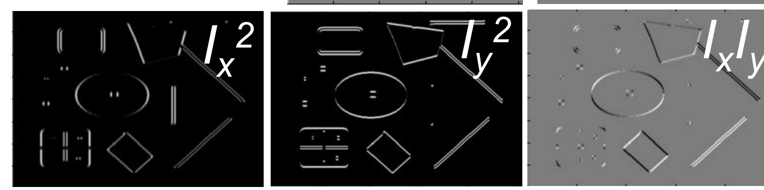
- Compute second moment matrix (autocorrelation matrix)

$$M(\sigma_I,\sigma_D)=g(\sigma_I)*\begin{bmatrix} I_x^2(\sigma_D) & I_xI_y(\sigma_D) \\ I_xI_y(\sigma_D) & I_y^2(\sigma_D) \end{bmatrix}$$

1. Image derivatives



2. Square of derivatives



3. Gaussian filter $g(\sigma_I)$

4. Cornerness function – two strong eigenvalues

$$R=\det[M(\sigma_I,\sigma_D)]-\alpha[\mathrm{trace}(M(\sigma_I,\sigma_D))]^2$$
$$=g(I_x^2)g(I_y^2)-[g(I_xI_y)]^2-\alpha[g(I_x^2)+g(I_y^2)]^2$$

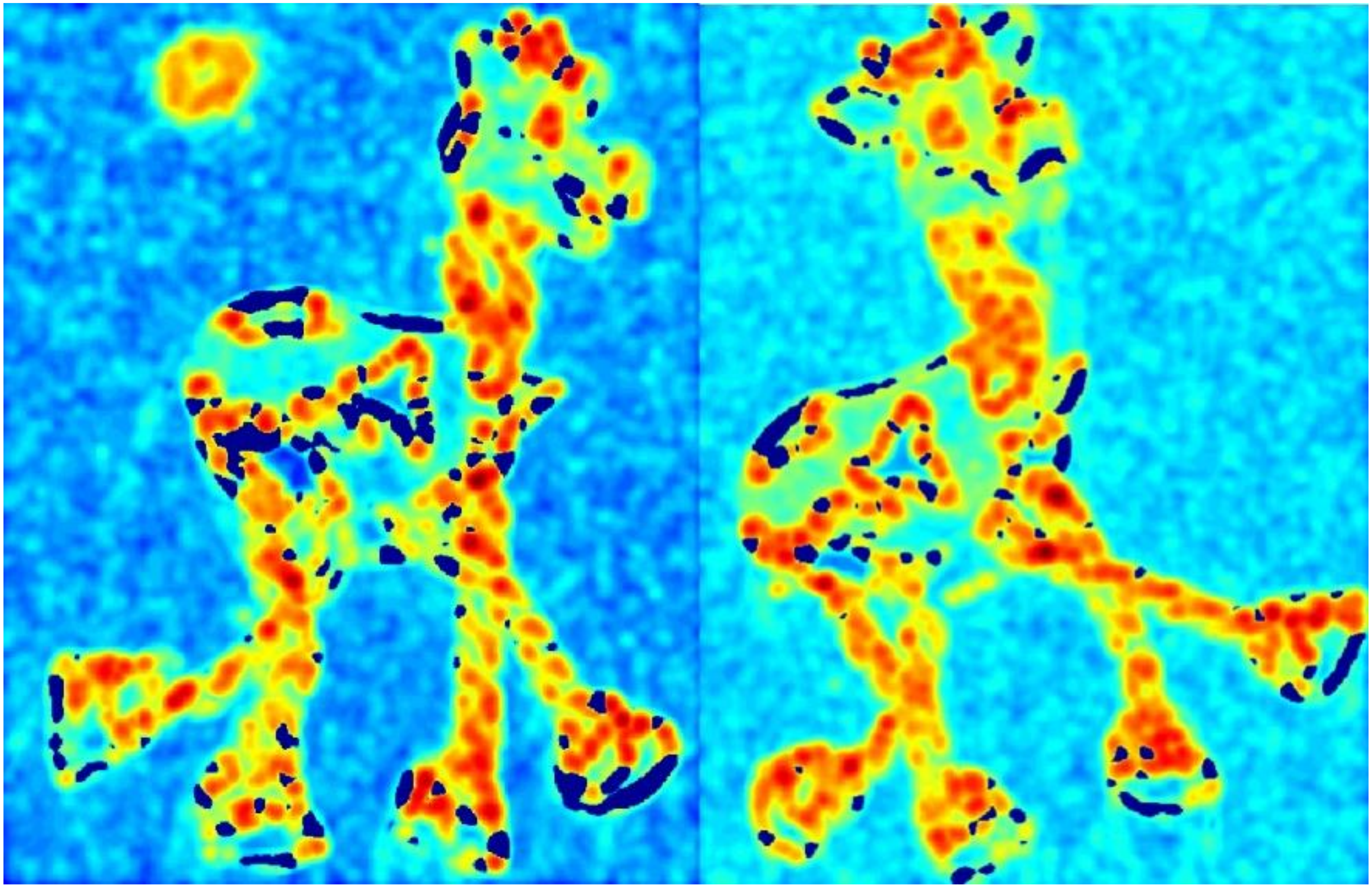
5. Perform non-maximum suppression

Slide credit: Krystian Mikolajczyk

Harris Detector: Workflow

Slide adapted from Darya Frolova, Denis Simakov

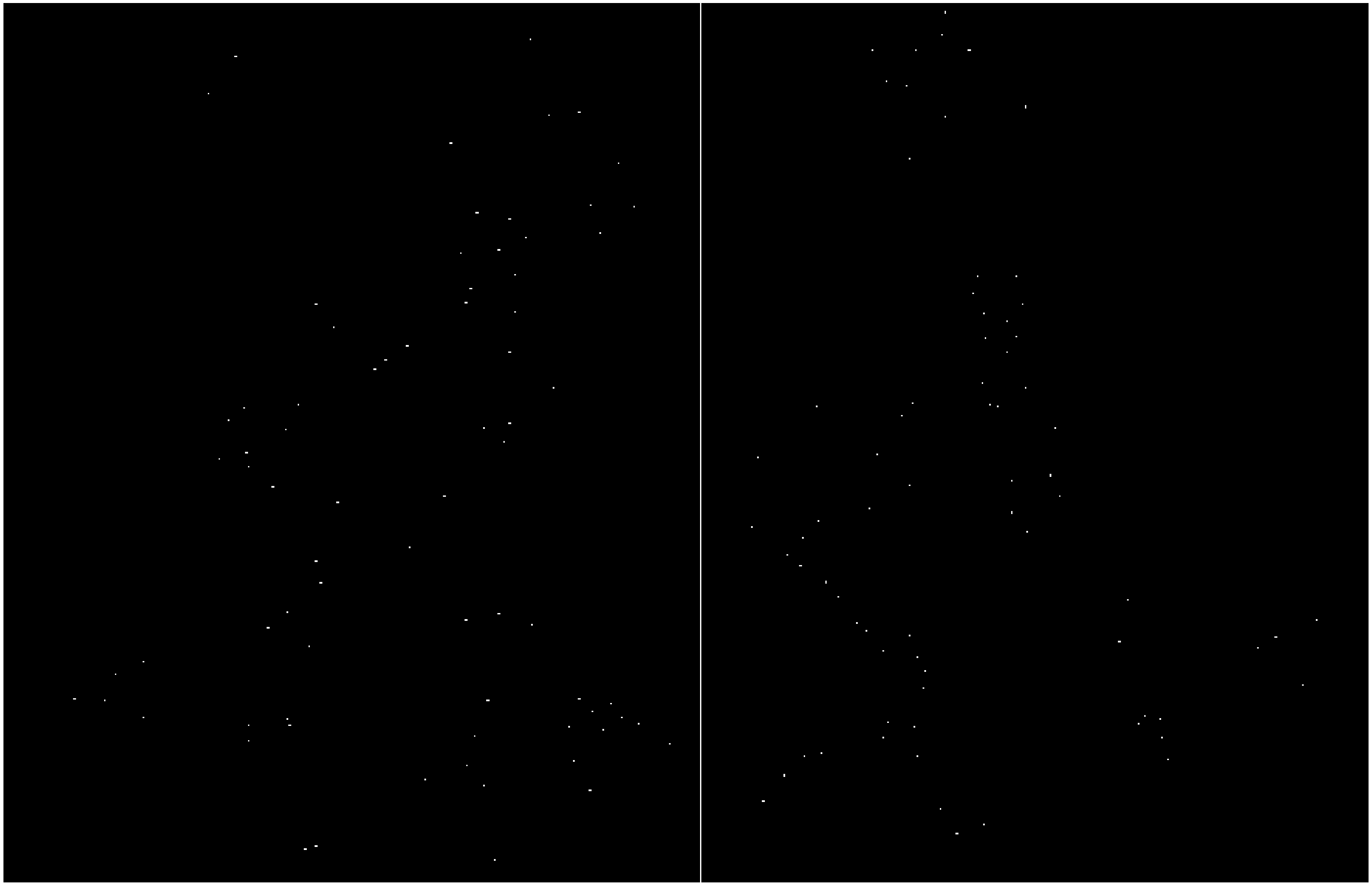
Harris Detector: Workflow



- Compute corner responses *R*

Slide adapted from Darya Frolova, Denis Simakov

Harris Detector: Workflow



- Take only the local maxima of R, where $R >$ threshold.

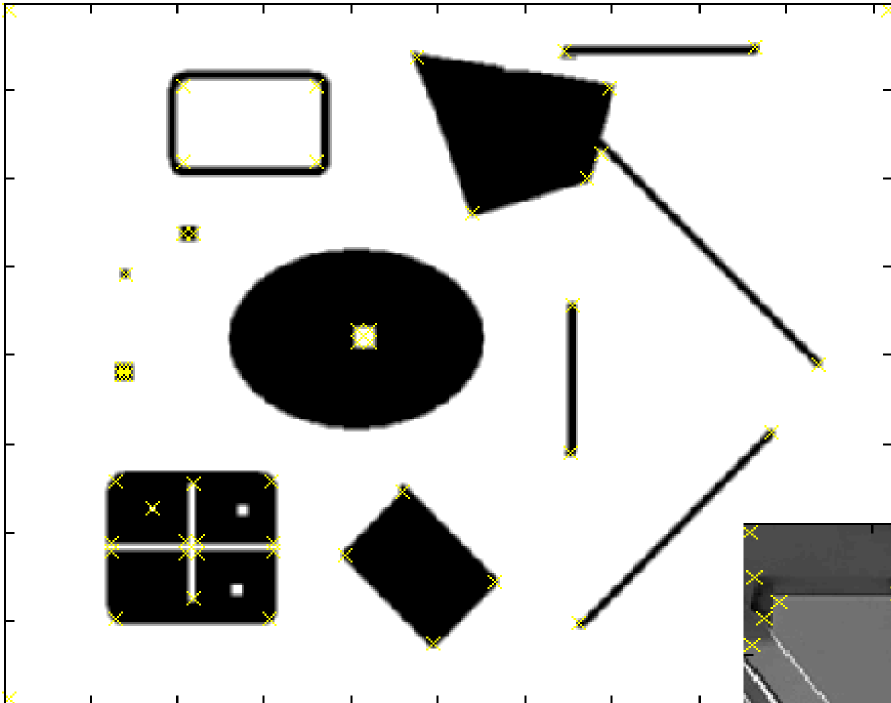
Slide adapted from Darya Frolova, Denis Simakov

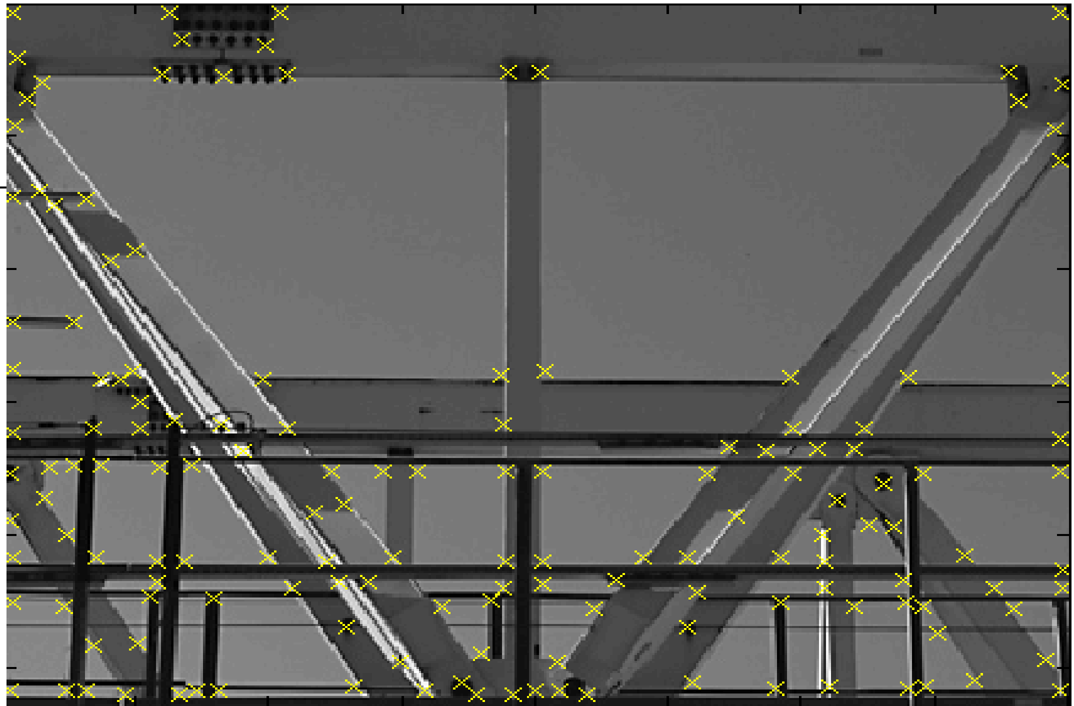
Harris Detector: Workflow

- Resulting Harris points

Slide adapted from Darya Frolova, Denis Simakov

Harris Detector – Responses [Harris88]

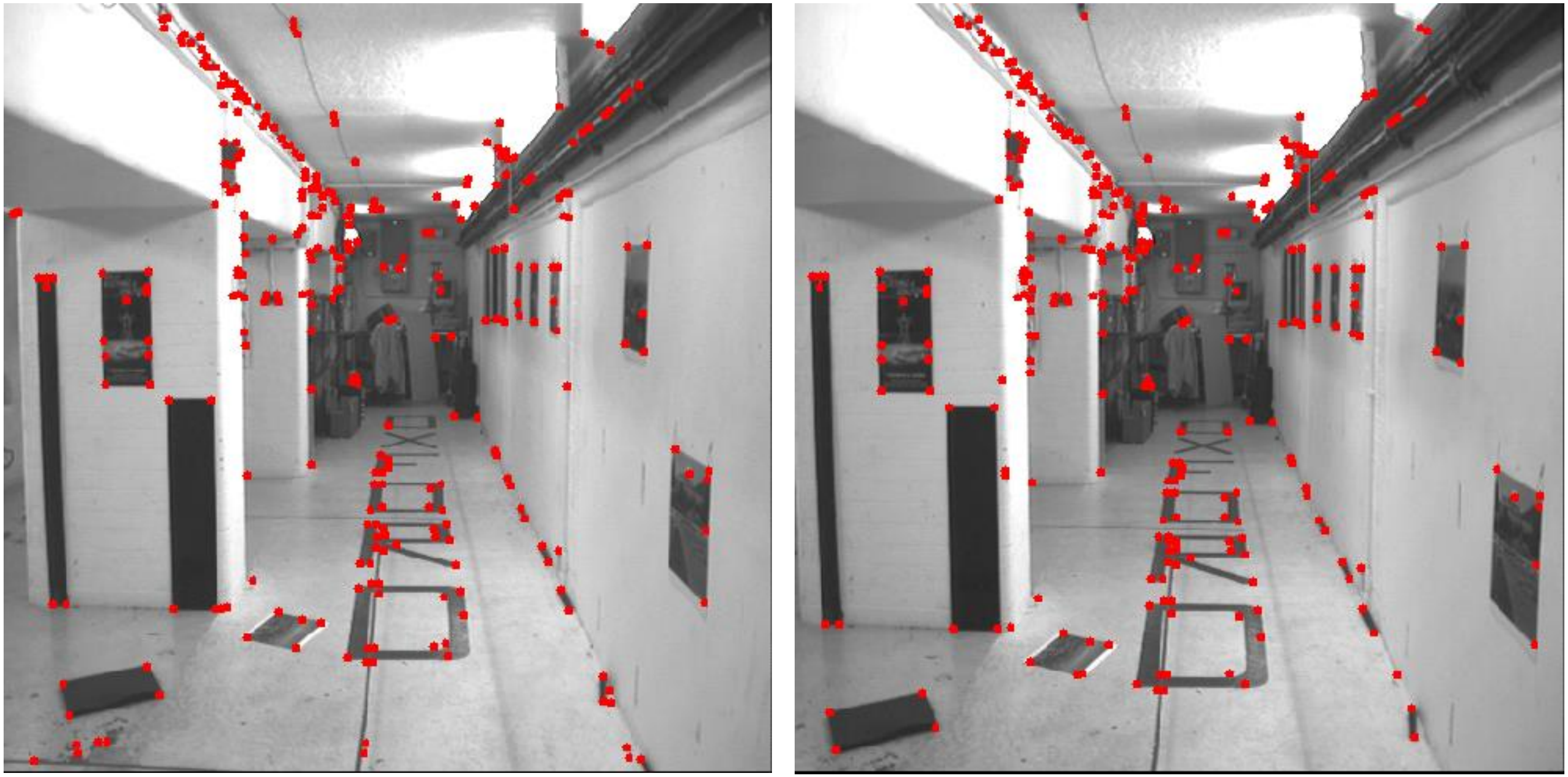




Effect: A very precise corner detector.

Slide credit: Krystian Mikolajczyk

Harris Detector – Responses [Harris88]

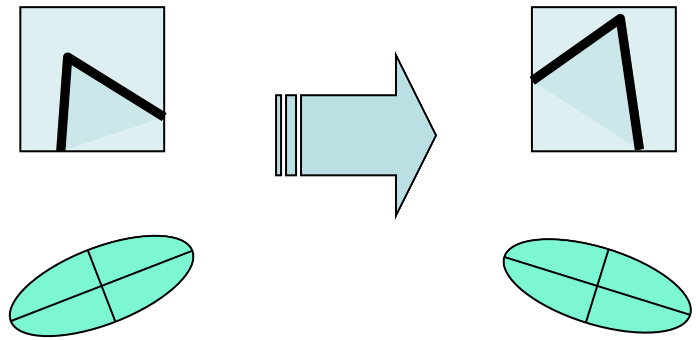


- Results are well suited for finding stereo correspondences

Slide credit: Kristen Grauman

Harris Detector: Properties

- Rotation invariance?



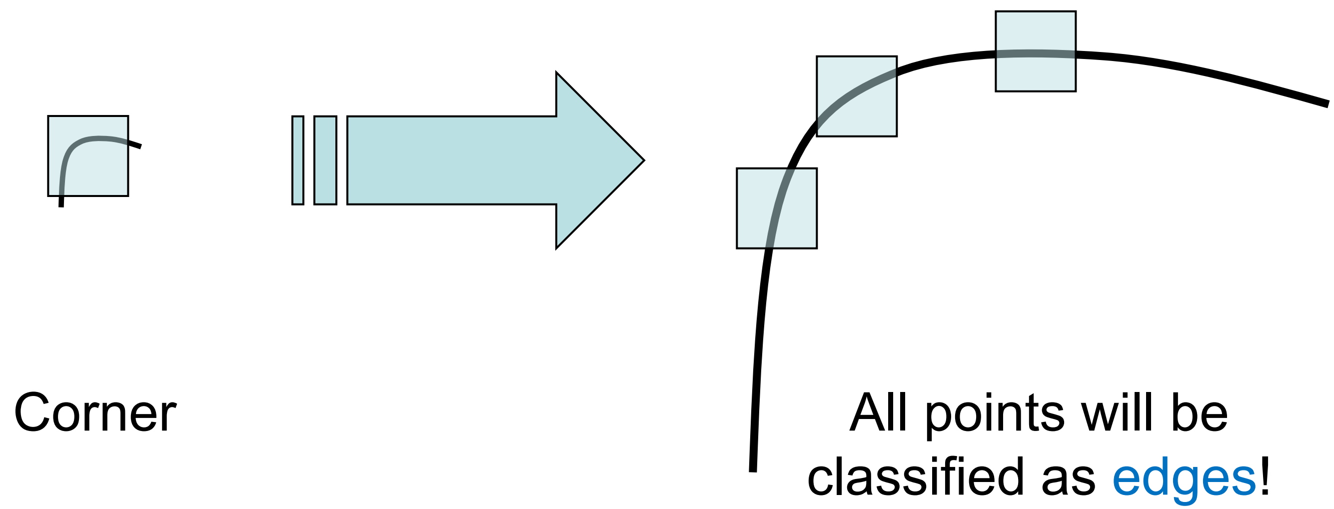
Ellipse rotates but its shape (i.e. eigenvalues) remains the same

Corner response R is invariant to image rotation

Slide credit: Kristen Grauman

Harris Detector: Properties

- Rotation invariance
- Scale invariance?



Not invariant to image scale!

Slide credit: Kristen Grauman

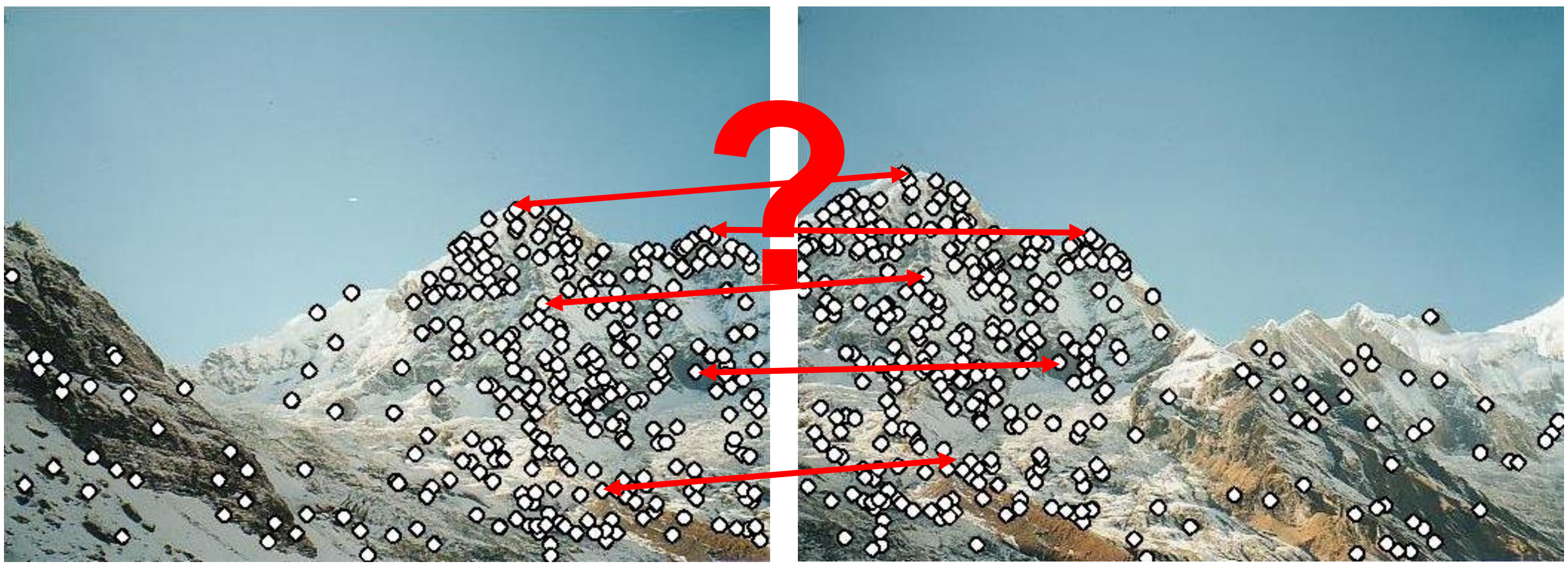
Topics of This Lecture

- 3D Reconstruction
  - Visual cues
  - Stereo vision
- Local Invariant Features
  - Motivation
  - Requirements, Invariances
- Keypoint Localization
  - Harris detector
- Local Descriptors
  - Orientation normalization
  - SIFT descriptor

Local Descriptors

- We know how to detect points
- Next question:

How to *describe* them for matching?



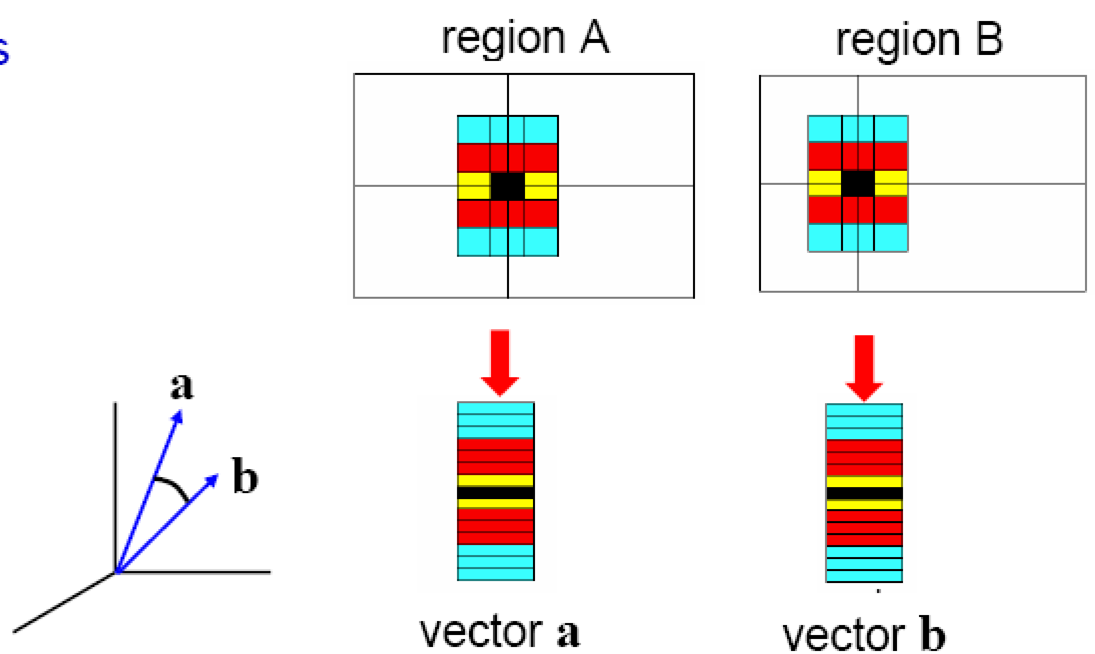
Point descriptor should be:
1. Invariant
2. Distinctive

Local Descriptors

- Simplest descriptor: list of intensities within a patch.
- What is this going to be invariant to?

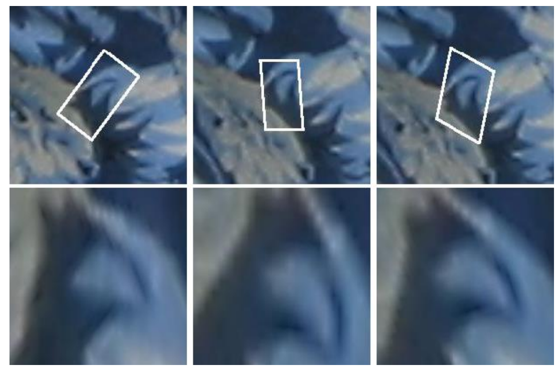
Write regions as vectors

$$\mathtt{A} \rightarrow \mathbf{a}, \;\; \mathtt{B} \rightarrow \mathbf{b}$$

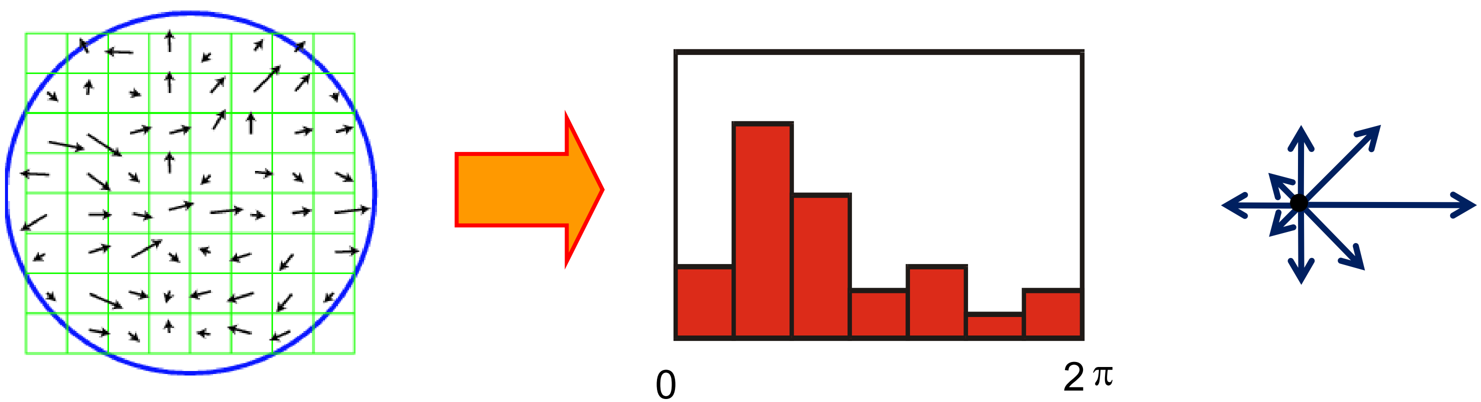


Feature Descriptors

- Disadvantage of patches as descriptors:
  - Small shifts can affect matching score a lot

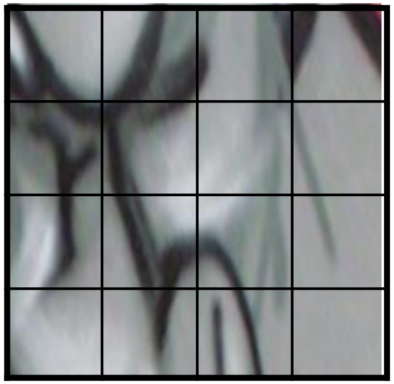


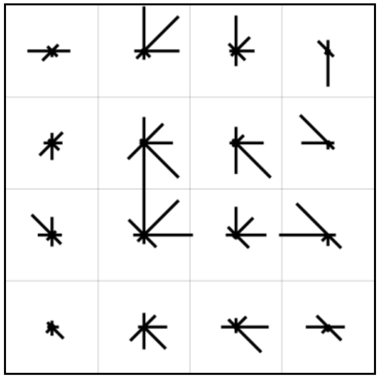
- Solution: histograms



Feature Descriptors: SIFT

- Scale Invariant Feature Transform
- Descriptor computation:
  - Divide patch into 4x4 sub-patches: 16 cells
  - Compute histogram of gradient orientations (8 reference angles) for all pixels inside each sub-patch
  - Resulting descriptor: 4x4x8 = 128 dimensions

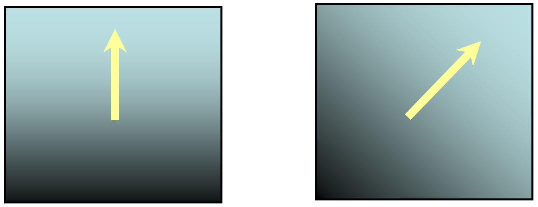




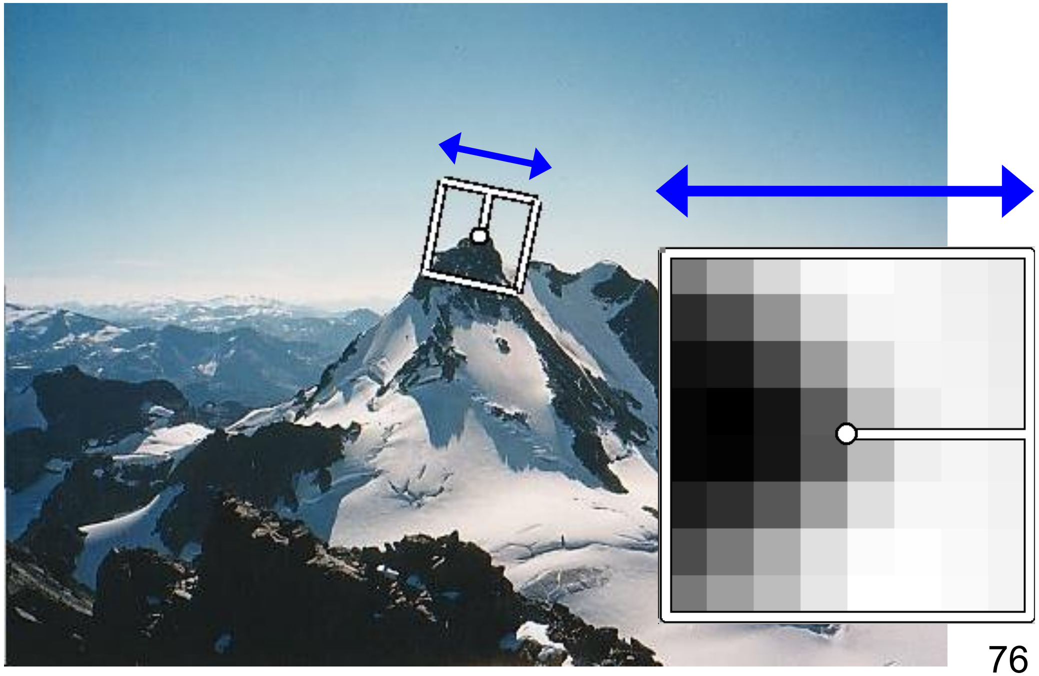
David G. Lowe. "Distinctive image features from scale-invariant keypoints." *IJCV* 60 (2), pp. 91-110, 2004.

Rotation Invariant Descriptors

- Find local orientation
  - Dominant direction of gradient for the image patch



- Rotate patch according to this angle
  - This puts the patches into a canonical orientation.

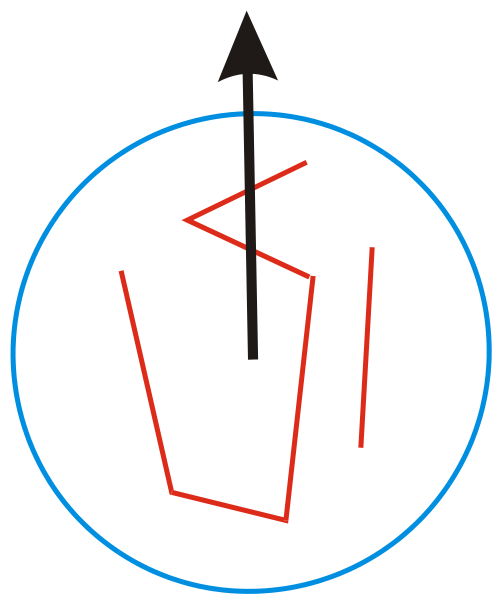


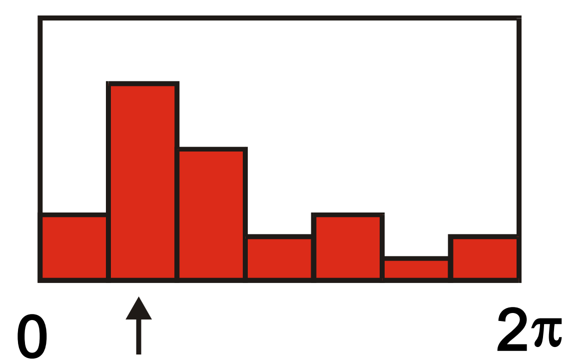
Slide credit: Svetlana Lazebnik, Matthew Brown

Orientation Normalization: Computation

- Compute orientation histogram
- Select dominant orientation
- Normalize: rotate to fixed orientation

[Lowe, 1999]

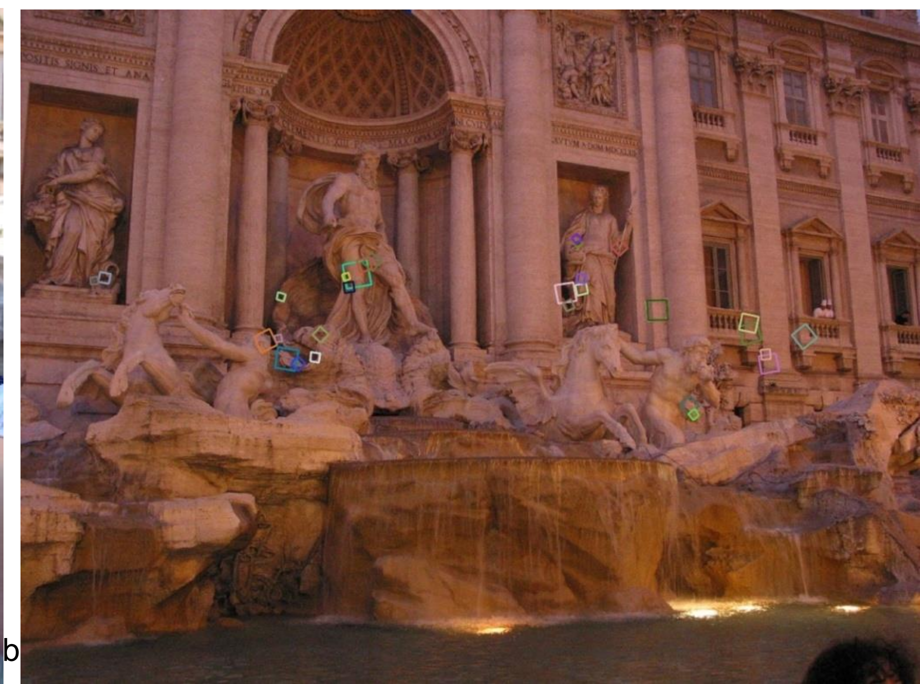




Slide adapted from David Lowe

Summary: SIFT

- Extraordinarily robust matching technique
  - Can handle changes in viewpoint up to ~60 deg. out-of-plane rotation
  - Can handle significant changes in illumination
    - Sometimes even day vs. night (below)
  - Fast and efficient—can run in real time
  - Lots of code available
    - http://people.csail.mit.edu/albert/ladypack/wiki/index.php/Known_implementations_of_SIFT



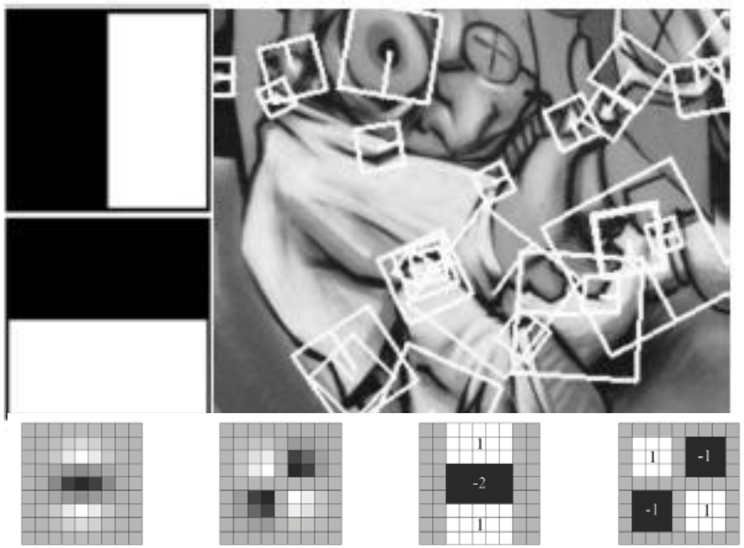
Slide credit: Steve Seitz

b

Working with SIFT Descriptors

- One image yields:
  - n 128-dimensional descriptors: each one is a histogram of the gradient orientations within a patch
    - [n x 128 matrix]
  - n scale parameters specifying the size of each patch
    - [n x 1 vector]
  - n orientation parameters specifying the angle of the patch
    - [n x 1 vector]
  - n 2D points giving positions of the patches
    - [n x 2 matrix]

Local Descriptors: SURF



- Fast approximation of SIFT idea
  - Efficient computation by 2D box filters & integral images
    $\Rightarrow$ 6 times faster than SIFT
  - Equivalent quality for object identification
  - http://www.vision.ee.ethz.ch/~surf
- GPU implementation available
  - Feature extraction @ 200Hz (detector + descriptor, 640×480 img)
  - http://homes.esat.kuleuven.be/~ncorneli/gpusurf/

[Bay, ECCV’06], [Cornelis, CVGPU’08]

You Can Try It At Home…

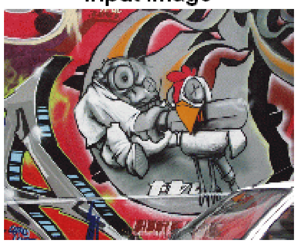
- For most local feature detectors, executables are available online:
- http://robots.ox.ac.uk/~vgg/research/affine
- http://www.cs.ubc.ca/~lowe/keypoints/
- http://www.vision.ee.ethz.ch/~surf
- http://homes.esat.kuleuven.be/~ncorneli/gpusurf/

Affine Covariant Features

Collaborative work between: the Visual Geometry Group, Katholieke Universiteit Leuven, Inria Rhone-Alpes and the Center for Machine Perception.

Affine Covariant Region Detectors

Input image



→ Region detector

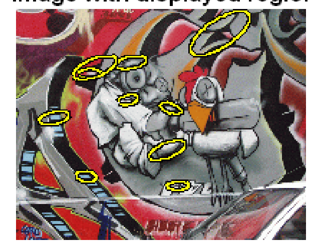
Detector output

format:
1.0
m
$u_1\ v_1\ a_1\ b_1\ c_1$
:
:
$u_m\ v_m\ a_m\ b_m\ c_m$

output example:
img1.haraff

→ display_features.m

Image with displayed regions



Parameters defining an affine region

`u,v,a,b,c` in `a(x-u)(x-u)+2b(x-u)(y-v)+c(y-v)(y-v)=1`
with `(0,0)` at image top left corner

Code

- provided by the authors, see publications for details and links to authors web sites.

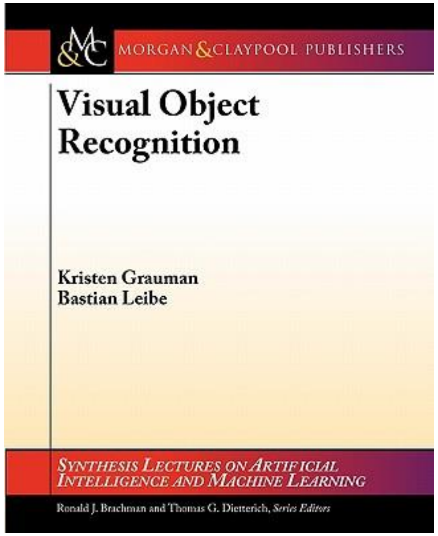
Linux binaries	Example of use	Displaying r
Harris-Affine & Hessian-Affine	`prompt>./h_affine.ln -haraff -i img1.ppm -o img1.haraff -thres 1000`	`matlab>> d`
	`prompt>./h_affine.ln -hesaff -i img1.ppm -o img1.hesaff -thres 500`	`matlab>> d`
MSER - Maximaly stable extremal regions (also Windows)	`prompt>./mser.ln -t 2 -es 2 -i img1.ppm -o img1.mser`	`matlab>> d`
IBR - Intensity extrema based detector	`prompt>./ibr.ln img1.ppm img1.ibr -scalefactor 1.0`	`matlab>> d`
EBR - Edge based detector	`prompt> ./ebr.ln img1.ppm img1.ebr`	`matlab>> d`
Salient region detector	`prompt>./salient.ln img1.ppm img1.sal`	`matlab>> d`

http://www.robots.ox.ac.uk/~vgg/research/affine/detectors.html#binaries

References and Further Reading

- Background information on local feature detectors and descriptors can be found in Chapter 3 of the following book.

  K. Grauman, B. Leibe
  Visual Object Recognition
  Morgan & Claypool publishers, 2011
  (available in the class moodle)



- Good survey paper on Int. Pt. detectors and descriptors
  - T. Tuytelaars, K. Mikolajczyk, *Local Invariant Feature Detectors: A Survey*, Foundations and Trends in Computer Graphics and Vision, Vol. 3, No. 3, pp 177-280, 2008.